



Recent development in ^1H NMR relaxometry

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Abstract

Nuclear Magnetic Resonance (NMR) relaxometry is one of the most powerful methods of studying dynamical properties of condensed matter and solids, from simple, small molecules and ions to complex, macromolecular systems. The unique advantages of NMR relaxometry are the possibility to probe dynamical processes across a broad range of correlation times (from ms to ns) in a single experiment and the ability to identify the mechanism of the molecular and ionic motion. In this chapter an overview of the large spectrum of possible applications of NMR relaxometry is presented in numerous examples. The examples include molecular and ionic liquids in bulk and confinement, nanofluids, proteins and polymers, complex solids, paramagnetic and superparamagnetic systems and they are accompanied by outlines of the aspects of spin relaxation theories that are most relevant for the specific classes of systems. The theory encompasses the subjects of several relaxation pathways and various spin interactions combined with advanced models of motion.

Keywords: NMR, Relaxation, Dynamics, Diffusion, Relaxation enhancement



1. Introduction

For nuclei of spin quantum number $I = \frac{1}{2}$ placed in an external magnetic field, $\vec{B}_0$, there are two available energy levels (two quantum states)—they are associated with the magnetic spin quantum numbers $m_I = \pm \frac{1}{2}$. The states are populated according to the Boltzmann distribution [1]. The difference in the populations leads to an effective magnetization, $\vec{M}_I$, parallel to $\vec{B}_0$. When the magnetic field changes to a new value, B_{rel} , the system re-populates evolving towards its new equilibrium determined by the Boltzmann distribution. The process is referred to as spin-lattice relaxation. In the simplest case the magnetization evolution is single exponential. For $B_{rel} < B_0$ the magnetization evolves in time following the equation: $M_I(t) = M_{I,0} \exp\left(-\frac{t}{T_1}\right)$, where $M_{I,0}$ is the initial magnetization (at the magnetic field of B_0), while T_1 is referred to as spin-lattice relaxation time; its reciprocal value, $R_1 = 1/T_1$, is called spin-lattice relaxation rate [2]. For $B_{rel} > B_0$ the time evolution of the magnetization can be described as: $M_I(t) = M_{I,0} \left(1 - \exp\left(-\frac{t}{T_1}\right)\right)$. The relaxation rate (time) depends on the magnetic field, B_{rel} . The magnetic field can be expressed in terms of the resonance frequency ν_I . The relationship between the two quantities yields: $\nu_I = \gamma_I B_{rel}$, where γ_I denotes the gyromagnetic factor of the nucleus of spin I . Therefore one can write: $R_1 = R_1(\omega_I)$, where $\omega_I = 2\pi\nu_I$. For $I = \frac{1}{2}$ the resonance frequency is equal to the energy difference (expressed in frequency units) between the two energy levels associated with the magnetic quantum numbers $m_I = \pm \frac{1}{2}$. For $I \geq 1$ the resonance frequency still corresponds to the energy difference between quantum states for which $|\Delta m_I| = 1$, but only under the assumption that the interaction of the nuclear spin with the external magnetic field (the Zeeman coupling) is the only interaction determining the energy level structure of the nucleus.

The re-population of the energy levels requires exchange of energy with the surrounding (referred to as lattice). The nuclear magnetic moments interact not only with the external magnetic field but also with each other. The mutual magnetic dipole-dipole interactions stochastically fluctuate in time as a result of molecular dynamics. Transitions between the spin energy levels are possible due to an exchange of energy between the spin system and the lattice.

In other words, the excess energy can be absorbed by the lattice; the lattice can also act as a source of energy if needed. The spin-lattice relaxation rate is determined by the probability of spin transitions between the energy levels. As the transitions are caused by the mutual magnetic dipole-dipole interactions, the transition probability depends on the strength (the amplitude) of the coupling. The second factor determining the transition probability is the time scale of the stochastic fluctuations of the dipole-dipole interactions [3–8]. When the time scale of the fluctuations corresponds to the resonance frequency, i.e., $\omega\tau_c \cong 1$, where τ_c denotes the characteristic time constant of the fluctuations (referred to as a correlation time), the transition probability reaches its maximum. The third, highly relevant factor is the mechanism of the fluctuations [3–20], i.e., the nature of the molecular motion: rotation, translation, vibration, etc. This statement can be somewhat surprising at the first sight. The effect is, however, directly related to the concept of a time correlation function. Time correlation functions describe how fast a system forgets its initial position. Let us consider as an example water molecule. The quantity of interest is, in this case, the orientation of the ^1H - ^1H magnetic dipole axis with respect to the direction of the external magnetic field, $\vec{B}_0$. The orientation stochastically fluctuates in time due to rotational motion of the molecule. Anticipating the content of the following sections, one can say at this moment that the corresponding correlation function is single-exponential (one should not confuse this statement with the single-exponential time evolution of the magnetization, mentioned above). As spin relaxation is represented in the frequency domain, Fourier transforms of time correlation functions are the exact quantities of interest. Fourier transform of a time correlation function is called a spectral density function. In the case of a single-exponential correlation function, the corresponding spectral density is of Lorentzian form. One should take into account that magnetic moments of ^1H nuclei belonging to different water molecules also interact with each other. In this case not only the orientation of the ^1H - ^1H dipole-dipole axis changes its orientation in time, but also the inter-nuclear distance. The dominating origin of the modulations is translation diffusion. The corresponding correlation function, and hence the spectral density function, is of much more complex form [9–20]. Spectral density functions can be considered as ‘histograms’ (distributions) of the frequencies of the motional processes. In other words, they represent the probability that the system performs motion of a given frequency. As the process of the re-population of the spin energy levels involves spin transitions of a

well-defined energy (frequency), the probability that this frequency is available in the spectrum of the molecular motion is of high importance for the transition to occur.

With this introductory picture of spin relaxation processes in mind, in the following sections a comprehensive overview of the spin relaxation theory is presented.



2. Relaxation theory for a two spin-1/2 system

Spin-lattice relaxation rate, $R_{1,I}(\omega_I)$, for two 1/2-spins, $I_1 = I_2 = \frac{1}{2}$, of the same resonance frequencies (for instance, two ^1H nuclei or two ^{19}F nuclei) is described by a very simple formula [3–8,21–24]:

$$R_{1,I}(\omega_I) = \frac{3}{10} \left(\frac{\mu_0 \gamma_I^2 \hbar}{4\pi r^3} \right)^2 \left[\frac{\tau_c}{1 + (\omega_I \tau_c)^2} + \frac{4\tau_c}{1 + (2\omega_I \tau_c)^2} \right] \quad (1)$$

where μ_0 and $\hbar$ denote the magnetic permeability and Planck constant, respectively, while r denotes the inter-spin distance. In Eq. (1) it has been assumed that the relaxation process is caused by magnetic dipole-dipole interactions and that the correlation function associated with the motion causing time fluctuations of the interactions is single exponential. As isotropic rotational diffusion is characterized by a single-exponential correlation function, Eq. (1) is typically used in connection to isotropic rotation. In this case one sets $\tau_c = \tau_{rot}$. It is important to stress at this stage that the correlation time associated with magnetic dipole-dipole interactions is a rank-two correlation time, as the quantum-mechanical representations of the dipole-dipole coupling includes rank-two tensor operators. Eq. (1) can be rewritten in the form:

$$\begin{aligned} R_{1,I}(\omega_I) &= 2I(I+1) \left(\frac{\mu_0 \gamma_I^2 \hbar}{4\pi r^3} \right)^2 [J_{rot}(\omega_I) + 4J_{rot}(2\omega_I)] \\ &= \frac{1}{3} I(I+1) a_{DD}^2 [J_{rot}(\omega_I) + 4J_{rot}(2\omega_I)] \end{aligned} \quad (2)$$

The rotational spectral density function, $J_{rot}(\omega)$, is defined as: $J_{rot}(\omega) = \frac{1}{5} \frac{\tau_{rot}}{1 + (\omega \tau_{rot})^2}$, while the dipole-dipole relaxation constant, a_{DD} , takes the form: $a_{DD} = \sqrt{6} \frac{\mu_0 \gamma_I^2 \hbar}{4\pi r^3}$. The coefficients ($\frac{1}{5}$ and $\sqrt{6}$) stem from a normalization of rank-two tensor operators.

The origin of the expression for $R_{1,I}(\omega_I)$ requires explanations. For a system of two 1/2-spins (I_1, I_2) there are four available quantum states denoted

as $|i\rangle = |m_1, m_2\rangle$: $|1\rangle = |\frac{1}{2}, \frac{1}{2}\rangle$, $|2\rangle = |-\frac{1}{2}, \frac{1}{2}\rangle$, $|3\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$, $|4\rangle = |-\frac{1}{2}, -\frac{1}{2}\rangle$. The magnetization can change as a result of the following spin transitions: $|1\rangle \rightarrow |2\rangle$, $|1\rangle \rightarrow |3\rangle$, $|2\rangle \rightarrow |4\rangle$, $|3\rangle \rightarrow |4\rangle$ and $|1\rangle \rightarrow |4\rangle$. The first four transitions are of the frequency ω_I , while the last one of $2\omega_I$. This explains the origin of the term $J(2\omega)$ in Eq. (1).

The quantum-mechanical representation of the magnetic dipole-dipole coupling takes the form [3–8,21–27]:

$$\begin{aligned} H_{DD}(I_1, I_2) &= a_{DD} \{ T_0^2(I_1, I_2) \} \\ &= \sqrt{6} \frac{\mu_0 \gamma_I^2 \hbar}{4\pi r^3} \left\{ \frac{1}{\sqrt{6}} \left[2I_1^z I_2^z - \frac{1}{2} (I_1^+ I_2^- + I_1^- I_2^+) \right] \right\} \quad (3) \end{aligned}$$

where $I^\pm = I^x \pm iI^y$ (I^x , I^y and I^z denote the components of the $\hat{I}$ operator); $H_{DD}(I_1, I_2)$ denotes the Hamiltonian of the magnetic dipole-dipole interaction. The Hamiltonian represents the dipole-dipole coupling in a molecular frame. The frame is fixed in the molecule and its z -axis is parallel to the inter-nuclear $I_1 - I_2$ axis. The dipole-dipole interaction is time independent from the perspective of the molecular frame. Moreover, the Hamiltonian includes only the $T_0^2(I_1, I_2)$ tensor component, while in order to induce the spin transitions, i.e., to change the total magnetic quantum number of the system, $\Delta(m_1 + m_2)$, by ± 1 (as required for the $|1\rangle \rightarrow |2\rangle$, $|1\rangle \rightarrow |3\rangle$, $|2\rangle \rightarrow |4\rangle$, $|3\rangle \rightarrow |4\rangle$ transitions) or ± 2 (the $|1\rangle \rightarrow |4\rangle$ transition), the Hamiltonian must include the $T_{\pm 1}^2(I_1, I_2)$ and $T_{\pm 2}^2(I_1, I_2)$ tensor component, respectively.

The fundamental point of spin relaxation theories is that the Hamiltonian of the interactions causing the relaxation process (the perturbing Hamiltonian) must be represented in the same reference frame as the Hamiltonian of the interaction determining the energy level structure of the spin system (the main Hamiltonian). The main Hamiltonian includes, in the present case, the Zeeman interaction of the nuclear magnetic moments with the external magnetic field $\vec{B}_0$; it yields: $H_Z(I_1, I_2) = \omega_I (\vec{I}_1^z + \vec{I}_2^z)$. The direction of the external magnetic field is chosen as the z -axis of the so called laboratory (L) frame. The Zeeman Hamiltonian $H_Z(I_1, I_2)$ is diagonal in this reference frame; i.e., the laboratory frame forms the principal axes system of the Zeeman coupling. This implies that the dipole-dipole Hamiltonian of Eq. (3) has to be transformed to the laboratory frame. This procedure seems to be quite cumbersome, but the transformation rules are well known. Before applying them let us generalize Eq. (3) to the form [4–7,24–27]:

$$H_{DD}(I_1, I_2) = a_{DD} \sum_{n=-2}^2 (-1)^n F_{-n}^2 T_n^2(I_1, I_2) \quad (4)$$

The tensor component $T_0^2(I_1, I_2)$ has already been defined in Eq. (3). The other tensor component yield: $T_{\pm 1}^2(I_1, I_2) = \mp \frac{1}{2} (I_1^\pm I_2^\pm + I_1^\pm I_2^\mp)$ and $T_{\pm 2}^2(I_1, I_2) = \frac{1}{2} I_1^\pm I_2^\pm$. Comparing Eq. (4) with Eq. (3) one can see that $F_0^2 = 1$, $F_{\pm 1}^2 = 0$, $F_{\pm 2}^2 = 0$. Then to get the laboratory frame representation of the dipole-dipole Hamiltonian one can straightforwardly apply the transformation [6,24–30]:

$$F_{-n}^{2(L)} = F_{-n}^{2(L)}(t) = \sum_{m=-2}^2 F_{-n}^2 D_{m,-n}^2(\Omega(t)) \quad (5)$$

The $D_{m,-n}^2(\Omega(t))$ quantities are called Wigner matrices. The name is somewhat misleading as they are just trigonometric functions of the set of Euler angles $\Omega(t) = (\alpha(t), \beta(t), \gamma(t))$ describing the orientation of the $I_1 - I_2$ dipole-dipole axis with respected to the direction of $\vec{B}_0$ [27–30]:

$$D_{0,0}^2(t) = \frac{3 \cos^2 \beta(t) - 1}{2} \quad (6a)$$

$$D_{0,\pm 1}^2(t) = \mp \sqrt{\frac{3}{2}} \sin \beta(t) \cos \beta(t) \exp(\pm i \gamma(t)) \quad (6b)$$

$$D_{0,\pm 2}^2(t) = \frac{1}{2} \sqrt{\frac{3}{2}} \sin^2 \beta(t) \exp(\pm 2i \gamma(t)) \quad (6c)$$

This implies that the dipole-dipole Hamiltonian takes in the laboratory frame the form:

$$H_{DD}^{(L)}(I_1, I_2, t) = a_{DD} \sum_{n=-2}^2 (-1)^n D_{0,-n}^2(t) T_n^2(I_1, I_2) \quad (7)$$

The Hamiltonian is now time dependent and includes all tensor components. Moreover, the Hamiltonian is expressed in terms of a product of time dependent quantities ($D_{0,-n}^2(t)$) and spin dependent quantities ($T_n^2(I_1, I_2)$).

The transition probabilities between two energy levels are proportional to a square of the matrix element of the dipole-dipole Hamiltonian represented in the basis constructed from the eigenfunctions of the Hamiltonian determining the energy level structure of the system; in the present case the $\{|i\rangle = |m_1, m_2\rangle\}$ basis. The matrix elements can be calculated using the formulae: $I^\pm |m_I\rangle = \sqrt{I(I+1) \mp m_I(m_I \pm 1)} |m_I \pm 1\rangle$, $I^\pm |m_I\rangle = m_I |m_I\rangle$ and keeping in mind the orthogonality properties of the $|m_1, m_2\rangle$ functions. To calculate the matrix element $\langle 1 | H_{DD}^{(L)}(I_1, I_2, t) | 2 \rangle$ it is sufficient to consider the part of the Hamiltonian for $n = -1$: $-a_{DD} D_{0,1}^2(t) T_{-1}^2(I_1, I_2)$; analogously

for $(1|H_{DD}^{(L)}(I_1, I_2, t)|4)$ one can take only the term for $n = -2$: $a_{DD}D_{0,2}^2(t)T_{-2}^2(I_1, I_2)$; all other terms give zero due to the orthogonality properties of the $|m_1, m_2\rangle$ functions. The matrix elements yield: $(1|H_{DD}^{(L)}(I_1, I_2, t)|2) = \frac{1}{4}a_{DD}D_{0,1}^2(t)$ and $(1|H_{DD}^{(L)}(I_1, I_2, t)|4) = \frac{1}{2}a_{DD}D_{0,2}^2(t)$. The ratio between the pre-factors of the spectral densities $J_{rot}(\omega_I)$ and $J_{rot}(2\omega_I)$ in Eq. (2), 1:4, corresponds to the ratio between the pre-factors of the $(1|H_{DD}^{(L)}(I_1, I_2, t)|2)^2$ and $(1|H_{DD}^{(L)}(I_1, I_2, t)|4)^2$ quantities, $(\frac{1}{4})^2 : (\frac{1}{2})^2$. The time dependent parts of the matrix elements (the Wigner matrices) form time correlation functions.

To get an intuitive understanding of this concept one can ask the following questions: the system evolves in time, so to which time point should one refer in the calculations? Moreover, thinking again about water molecules as an example, how to take into account that at the same time point the orientation of the $I_1 - I_2$ dipole-dipole axis is different for different molecules? A time correlation function $C(t)$ for a quantity $A(x(t))$ is defined as [3–8,30–33]:

$$C(t) = \overline{A^*(t)A(0)} = \iint A^*(x(t))A(x(0))P(x(t), x_0(0), t)P_{eq}(x_0)dx_0dx \quad (8)$$

where $P(x(t), x_0(0), t)$ denotes the probability that the system is at the position x at time t , provided it has been at the position x_0 at time zero, while $P_{eq}(x_0)$ is the equilibrium distribution of the positions of the system; $*$ denotes complex conjugation. The integral represents an averaging over all available initial and final (after the time t) positions of the system. For isotropic molecular tumbling the correlation function is, as already mentioned, single exponential [3–8]:

$$C_{rot,iso}(t) = \overline{D_{0,n}^{2*}(t)D_{0,n}^2(0)} = \frac{1}{5} \exp\left(-\frac{t}{\tau_{rot}}\right) \quad (9)$$

and independent of n . Fourier transform of the correlation function leads to the spectral density functions: $J_{rot}(\omega) = \frac{1}{5} \frac{\tau_{rot}}{1 + (\omega\tau_{rot})^2} = \text{Re} \int_{-\infty}^{\infty} C_{rot,iso}(t) \exp(-i\omega t) dt$.

Fig. 1 shows examples of simulated ^1H spin-lattice relaxation rates, $R_{1,I}(\omega_I) = R_{1H}(\omega_H)$, (ω_H denotes ^1H resonance frequency) depending on the rotational correlation time, τ_{rot} .

Real molecules are rarely spherical. Consequently, isotropic molecular tumbling is rather an exception than a rule. There are closed form

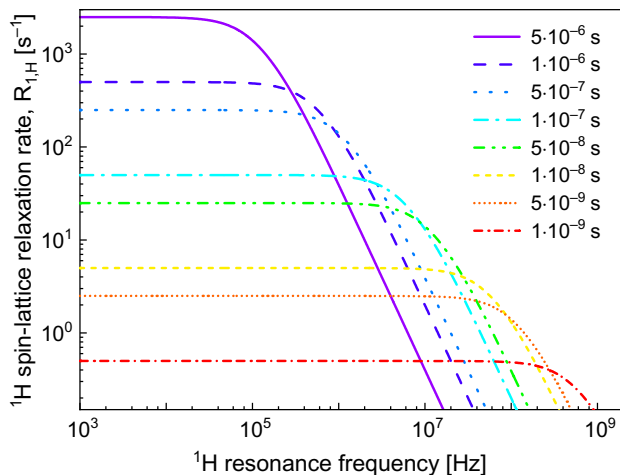


Fig. 1 ^1H spin-lattice relaxation rates simulated in terms of Eq. (1) and (2) for different values of the correlation time $\tau_c = \tau_{rot}$.

expressions for spectral density functions characterizing anisotropic molecular tumbling; an example is given by Eq. (1) [5,8,34,35]:

$$J_{rot,aniso}(\omega) = \frac{1}{5} \left[\frac{(3 \cos^2 \theta - 1)^2}{4} \frac{6D_{\perp}}{(6D_{\perp})^2 + \omega^2} + 3 \cos^2 \theta \sin^2 \theta \frac{5D_{\perp} + D_{\parallel}}{(5D_{\perp} + D_{\parallel})^2 + \omega^2} + \frac{3}{4} \sin^4 \theta \frac{2D_{\perp} + 4D_{\parallel}}{(2D_{\perp} + 4D_{\parallel})^2 + \omega^2} \right] \quad (10)$$

The angle θ is the angle between the $I_1 - I_2$ dipole-dipole axis and the rotation axis, while the diffusion coefficients, $D_{\perp}$ and $D_{\parallel}$, are related to the components of the rotational diffusion tensor: $D_{\perp} = D_{xx} = D_{yy}$, $D_{\parallel} = D_{zz}$. Eq. (10) can be expressed in terms of rotational correlation times defined as: $\frac{1}{\tau_{rot,n}} = 6D_{\perp} + n^2(D_{\parallel} - D_{\perp})$:

$$J_{rot,aniso}(\omega) = \frac{1}{5} \left[\frac{(3 \cos^2 \theta - 1)^2}{4} \frac{\tau_{2,0}}{1 + (\omega \tau_{2,0})^2} + 3 \cos^2 \theta \sin^2 \theta \frac{\tau_{2,1}}{1 + (\omega \tau_{2,1})^2} + \frac{3}{4} \sin^4 \theta \frac{\tau_{2,2}}{1 + (\omega \tau_{2,2})^2} \right] \quad (11)$$

One should note that the pre-factors of the Lorentzian functions of Eq. (11) are related to the Wigner rotation matrices: $D_{0,0}^{2*}(0, \theta, 0)$
 $D_{0,0}^2(0, \theta, 0) = \left(\frac{3\cos^2\theta - 1}{2}\right)^2$, $D_{0,1}^{2*}(0, \theta, 0)D_{0,1}^2(0, \theta, 0) = D_{0,-1}^{2*}(0, \theta, 0)D_{0,-1}^2(0, \theta, 0) = \frac{3}{2}\cos^2\theta\sin^2\theta$ and $D_{0,2}^{2*}(0, \theta, 0)D_{0,2}^2(0, \theta, 0) = D_{0,-2}^{2*}(0, \theta, 0)D_{0,-2}^2(0, \theta, 0) = \frac{3}{8}\sin^4\theta$.

Unfortunately (in the opinion of the authors), deviations from the simple Lorentzian form of rotational correlation function with a single correlation time are often accounted for by introducing phenomenological functions adopted from dielectric relaxation. Dielectric spectroscopy is based on the interaction of an external electric field with the [electric dipole moment](#) of the molecule. Dielectric relaxation spectroscopy does not allow to inquire into the molecular dynamics on an atomistic level—the method traces the reorientation of the overall electric dipole moment [36]. To be more specific, one can say that for small molecules, dielectric spectroscopy probes the overall molecular rotation. For more complex (larger) molecules, like, for example, polymers the local segmental dynamics (provided the dipole moment is transverse to the polymer chain) or the backbone reorientation (for polymers for which there is a component of their electric dipole moment parallel to the backbone) is probed [17,37,38]. The rotational correlation time determined by means of dielectric spectroscopy is rank-one correlation time, $\tau_{rot}^{(1)}$, while, as already mentioned, NMR relaxation is associated with rank-two correlation time, $\tau_{rot}^{(2)}$ [17,39]. The index (2) has been omitted in the in the above provided equations, for simplicity. The relationship between the two quantities yields: $\tau_{rot}^{(1)} = 3\tau_{rot}^{(2)}$, provided the rotational dynamics is a small step process [1,17,39].

Dielectric relaxation is represented in terms of susceptibility functions, $\chi(\omega)$, defined as: $\chi(\omega) = \omega J(\omega)$. For the Lorentzian spectral density, the susceptibility function takes the Debye form: $\chi_{Debye}(\omega) = \frac{\omega\tau_{rot}^{(1)}}{1 + (\omega\tau_{rot}^{(1)})^2}$. One of the most frequently used forms of susceptibility, and hence spectral density, functions is the Cole–Davidson function [17]:

$$\chi_{CD}(\omega) = \omega J_{CD}(\omega) = \frac{\sin \left[\beta \arctan \left(\omega\tau_{rot}^{(1)} / \beta \right) \right]}{\left[1 + \left(\omega\tau_{rot}^{(1)} / \beta \right)^2 \right]^{\beta/2}} \quad (11)$$

The coefficient β is in range $0 < \beta \leq 1$; for $\beta = 1$ Eq. (11) converges to the Debye form. As far as the physical meaning of the Cole–Davidson function is concerned, it can be in a good approximation treated as Fourier transform of

a stretched exponential correlation function: $C(t) = \exp \left[-\left(\frac{t}{\tau_{CD}} \right)^\beta \right]$.

A stretched exponential correlation function has been theoretically predicted for liquids of high viscosity [40–43]—the stretching is considered as a measure of cooperativity of the molecular dynamics resulted from strong intermolecular interactions (the average distance between them exceeds the molecular size by not more than 30%). The dielectric relaxation peaks for highly viscous liquids and polymers are indeed invariably broader than a Debye function, making this concept highly convincing. One should, however, be aware that from the physical point of view the Cole–Davidson spectral density implies heterogeneous rotational dynamics with a distribution of correlation times extended towards slow dynamics. For many systems such a motional scenario is hardly justified. Following this line, the Havrila–Negami susceptibility function [36], $\chi_{HN}(\omega)$, includes even two additional, phenomenological parameters (α and β) compared to the Debye function:

$$\begin{aligned} \chi_{HN}(\omega) &= \omega J_{HN}(\omega) \\ &= \left[1 + 2 \left(\omega \tau_{rot}^{(1)} \right)^\alpha \cos \left(\pi \frac{\alpha}{2} \right) + \left(\omega \tau_{rot}^{(1)} \right)^{2\alpha} \right]^{-\frac{\beta}{2}} \\ &\quad * \sin \left(\beta \arctan \frac{\left(\omega \tau_{rot}^{(1)} \right)^\alpha \sin \left(\pi \frac{\alpha}{2} \right)}{1 + \left(\omega \tau_{rot}^{(1)} \right)^\alpha \cos \left(\pi \frac{\alpha}{2} \right)} \right) \end{aligned} \quad (12)$$

One can associate the Havrila–Negami function with a distribution of correlation times (in analogy to the Cole–Davidson function), but the physical origin of such a distribution is not clear. Another frequently used dielectric function, Cole–Cole function, does not converge in the low frequency range when one tries to divide it frequency to use the density function representation. This makes it unsuitable for NMR relaxometry even from the mathematical viewpoint. The functions are compared in Fig. 2.

The subject of the physical models hidden behind the function typically used in Dielectric Spectroscopy has been addressed in [44] in the context of solutions of the protein bovine pancreatic trypsin inhibitor. It has been shown that the presence salts in the solutions leads to slowing down of protein rotation with little change of protein structure as it might be concluded from applying the commonly used Dielectric Spectroscopy functions. The finding strongly indicates that functions are (at least in some cases) unphysical and the parameters do not have the physical meaning usually attributed to them. The work has been rising serious controversies; in our opinion but the

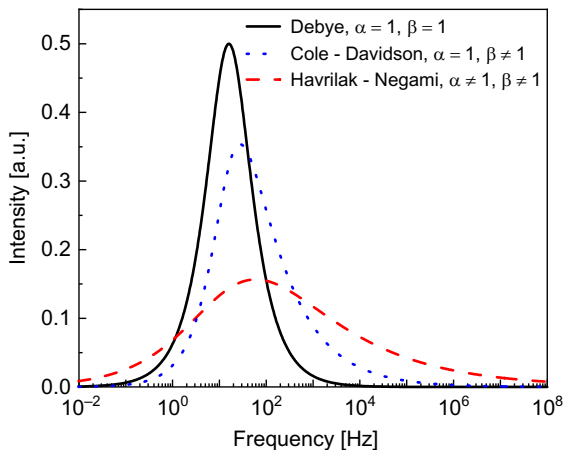


Fig. 2 Comparison of the shapes of Havriliak–Negami function for $\alpha=0.5$ and $\beta=0.5$, Cole–Davidson function that corresponds to Havriliak–Negami function for $\alpha=1$ and Debye function corresponding to the Cole–Davidson function for $\beta=1$.

mathematical arguments invoked in the paper are convincing. The Cole–Davidson distribution of the correlation times has been tested against a concept of two correlation times characterizing the rotation-like dynamics of pyridinium cations ($\text{C}_5\text{H}_6\text{N}$) in the molecular crystal $(\text{C}_5\text{H}_6\text{N})_5\text{Bi}_2\text{Br}_{11}$ [45]. It has turned out that the model assuming two correlation times differing by about one order of magnitude and attributed to the reorientation of the cation around its perpendicular axes leads to a very good agreement with the experimental data, as shown in Fig. 3.

The correlation function for translation diffusion, $C_{\text{trans}}(t)$, includes time fluctuations of not only the $I_1 - I_2$ dipole-dipole axis, but also of the inter-spin (inter-nuclear) distance, $r(t)$:

$$C_{\text{trans}}(t) = \frac{\overline{D_{0,n}^{2*}(t) D_{0,n}^2(0)}}{r(t)^3 r(0)^3} \quad (13)$$

For 3D (three dimensional) force-free diffusion with no geometrical restrictions the corresponding spectral density function takes the form [9,11,15]:

$$\begin{aligned} J_{\text{trans},(3D)}(\omega) &= \frac{72}{5} \frac{1}{d^3} N \left[\int_0^8 \frac{u^4}{81 + 9u^2 - 2u^4 + u^6} \exp\left(-\frac{u^2 t}{\tau_{\text{trans}}}\right) du \right] \exp(-i\omega t) dt \\ &= \frac{72}{5} \frac{1}{d^3} N \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \frac{u^2 \tau_{\text{trans}}}{u^4 + (\omega \tau_{\text{trans}})^2} du \end{aligned} \quad (14)$$

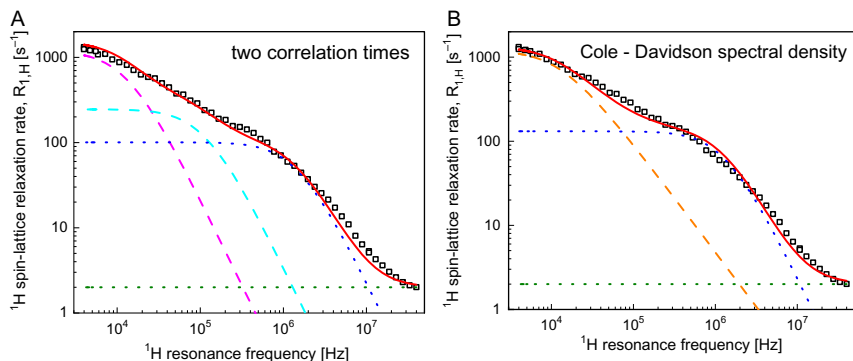


Fig. 3 ^1H spin-lattice relaxation data for $(\text{C}_5\text{H}_6\text{N})_5\text{Bi}_2\text{Br}_{11}$ at 273 K reproduced in terms of motional models assuming two correlation times for the dynamics of pyridinium cations (A) and a Cole–Davidson spectral density (B); solid red line—the overall fit, magenta and light blue dashed line (B)—relaxation contributions associated with rotation-like dynamics of pyridinium cations around its perpendicular axes replaced in (B) by a relaxation contribution modelled in terms of a Cole–Davidson spectral density—dashed orange line; blue and green dotted lines—additional relaxation contributions associated with faster dynamics. Reprinted from Ref. D. Kruk, M. Florek-Wojciechowska, R. Jakubas, S.K. Chaurasia, Sz. Brym, *Dynamics of molecular crystals by means of ^1H NMR relaxometry: dynamical heterogeneity versus homogenous motion*, *ChemPhysChem* 17 (2016) 2329–39, with permission.

The correlation time τ_{trans} is defined as: $\tau_{\text{trans}} = \frac{d^2}{D_{12}}$, where D_{12} denotes a relative translation diffusion coefficient being a sum of the self-diffusion coefficients of the interacting molecules: $D_{12} = D_{12}$ (for identical molecules $D_1 = D_2 = D_{\text{trans}}$, and thus $D_{12} = 2D_{\text{trans}}$). The parameter d is referred to as the distance of closest approach and it can be approximated by a sum of the radii of the molecules, while N denotes the number of the I -spin nuclei per unit volume. The integral in Eq. (14) can be approximated by the expression [5]:

$$J_{\text{trans},(3D)}(\omega) = \frac{72}{5} \frac{1}{d^3} N \frac{1 + \frac{5z}{8} + \frac{z^2}{8}}{1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{4z^4}{81} + \frac{z^5}{81} + \frac{z^6}{648}}, z = \sqrt{\omega\tau_{\text{trans}}} \quad (15)$$

The form of the spectral density function for translation diffusion is a fingerprint of the dimensionality of the diffusion process. For two dimensional (2D) diffusion, the spectral density function yields [13]:

$$J_{\text{trans},(2D)}(\omega) \propto \tau_{\text{trans}} \ln \left[1 + \frac{1}{(\omega\tau_{\text{trans}})^2} \right] \quad (16)$$

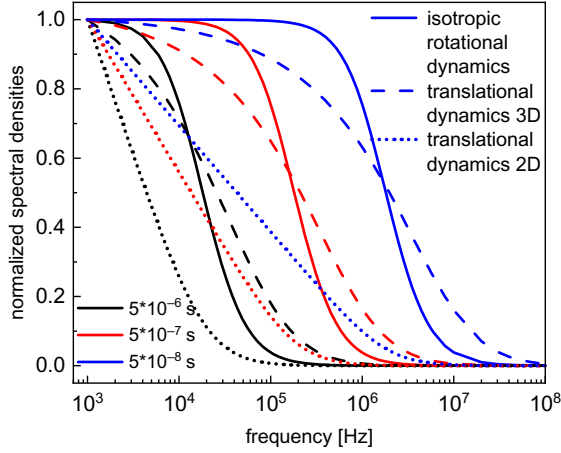


Fig. 4 Shapes of rotational and translational spectral density functions for $\tau_{\text{rot}} = \tau_{\text{trans}} = 5 \cdot 10^{-6} \text{ s}$, $5 \cdot 10^{-7} \text{ s}$ and $5 \cdot 10^{-8} \text{ s}$.

Fig. 4 shows a comparison between the shapes of the spectral density functions for isotropic rotation, 3D translation diffusion and 2D translation diffusion. The shapes are considerably different.

It is of interest to analyse the shape of the spectral density functions for translation diffusion in the low frequency range for which the condition $\omega\tau_{\text{trans}} \ll 1$ is fulfilled. For the case of 3D diffusion one obtains [13,16]:

$$J_{\text{trans,(3D)}}(\omega) \cong a - b\sqrt{\omega}, \quad b = \frac{2^{3/2}\pi}{45D_{12}^{3/2}} \quad (17)$$

This low frequency approximation implies that the spin-lattice relaxation rate $R_{1,I}(\omega_I)$ associated with 3D translation diffusion becomes a linear function of $\sqrt{\omega_I}$ [16]:

$$\begin{aligned} R_{1,I}(\omega_I) &\cong R_{1,I}(0) - B\sqrt{\omega_I} \\ &= R_{1,I}(0) - \frac{\sqrt{2}\pi}{15} \left(1 + 4\sqrt{2}\right) \left(\frac{\mu_0}{4\pi} \gamma_I^2 \hbar\right)^2 N D_{12}^{-3/2} \sqrt{\omega_I} \end{aligned} \quad (18)$$

The relationship allows to straightforwardly determine the translation diffusion coefficient from the low frequency slope of $R_{1,I}(\omega_I)$ plotted vs $\sqrt{\omega_I}$ as shown in Fig. 5 [16,46].

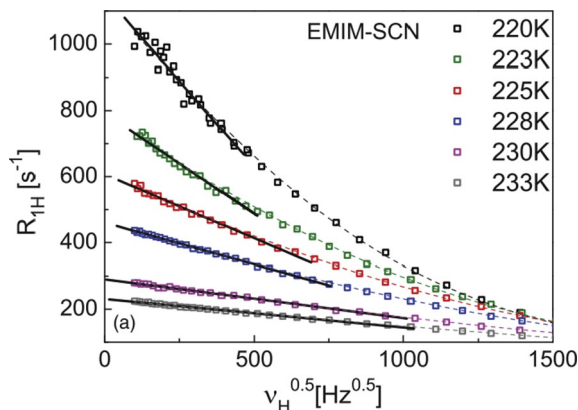


Fig. 5 ^1H spin-lattice relaxation rates, $R_{1,H}(\omega_H)$, for ionic liquid EMIM-SCN vs square root of the resonance frequency. Solid lines indicate the low frequency ranges of linear dependencies. Reprinted from Ref. D. Kruk, R. Meier, A. Rachocki, A. Korpala, R.K. Singh, E.A. Rossler, *Determining diffusion coefficients of ionic liquids by means of field cycling nuclear magnetic resonance relaxometry*, *J. Chem. Phys.* 140 (2014) 244509, with the permission of AIP Publishing.

One can also determine the asymptotic behaviour (for $\omega\tau_{\text{trans}} \ll 1$) of the translational spectral densities for the case of 2D and 1D diffusion; the corresponding expressions yield [13,14]:

$$J_{\text{trans},(2D)}(\omega) \propto [-\tau_{\text{trans}} \ln(\omega\tau_{\text{trans}})] \quad (19)$$

$$J_{\text{trans},(1D)}(\omega) \propto \frac{1}{\sqrt{\omega\tau_{\text{trans}}}} \quad (20)$$

For 1D diffusion there is no closed form expression available beyond the low frequency range.

The theoretically predicted linear dependence of $R_{1,I}(\omega_I)$ on $\frac{1}{\sqrt{\omega_I}}$ can be experimentally confirmed for liquids placed in nanotubes that provide 1D confinement. Fig. 6 shows a comparison between the shapes of $R_{1,H}(\omega_H)$ for BMIM-PF₆ in bulk and confined in carbon nanotubes. The linear dependence on $\frac{1}{\sqrt{\omega_H}}$ is clearly seen for the last case.

Combining the Arrhenius dependency of the correlation times on temperature [1], $\tau_c(T) = \tau_0 \exp\left(-\frac{E_A}{RT}\right)$ (where τ_0 can be considered as the high temperature limit of τ_c , while E_A denotes the activation energy of the motional process) with the expressions for the motion-specific spectral density functions, one can predict interesting properties of the relaxation rates measured vs temperature, depending on the mechanism of motion.

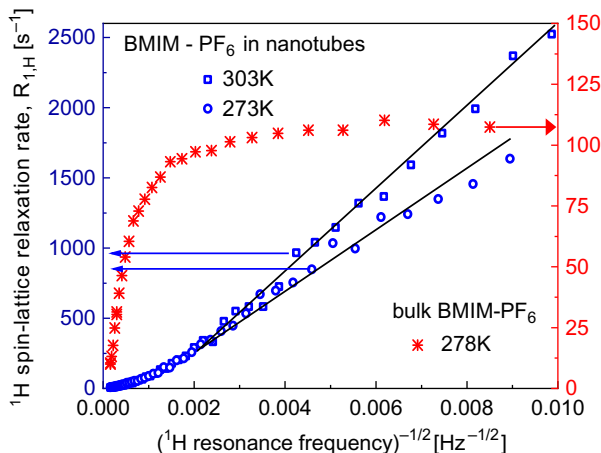


Fig. 6 ^1H spin-lattice relaxation rates, $R_{1,H}(\omega_H)$, for BMIM- PF_6 in bulk and confined in carbon nanotubes.

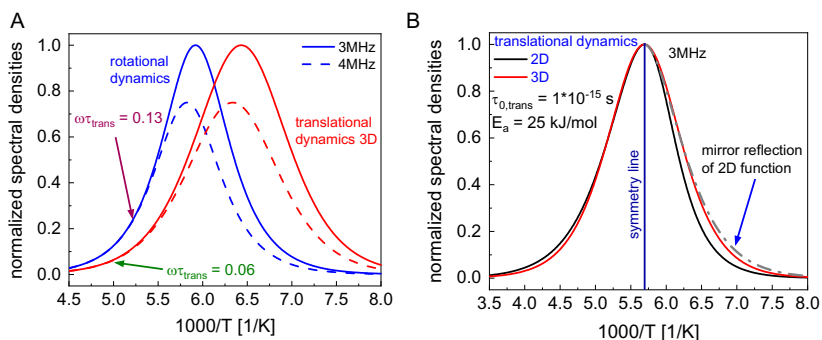


Fig. 7 Comparison of the shape of the spectral density functions for (A) isotropic rotational dynamics and 3D translation diffusion; (B) 2D and 3D translation diffusion vs reciprocal temperature. Asymmetry of the spectral density function for 2D translation motion is indicated.

One can, for instance, conclude that for isotropic rotational dynamics the limit of $\omega\tau_c$ for which the relaxation rates become frequency independent is reached earlier than for translation dynamics (Fig. 7A). Another very interesting observation is that the slope of the relaxation rate plotted vs reciprocal temperature is asymmetric for 2D diffusion in contrary to the case of 3D diffusion (Fig. 7B).

These properties should be considered parallel to the shape of the frequency dependencies of the spin-lattice relaxation rates to strengthen the conclusions regarding the mechanism of the molecular motion.



3. Quadrupole relaxation enhancement

As already explained, the relaxation is most effective when the correlation time of the motional process causing the relaxation process matches the resonance frequency. This implies that in relaxation experiments performed in the frequency range from about 1 kHz to 120 MHz (referring to the ^1H resonance frequency) one can probe molecular dynamics of the time scale from 10^{-3} to 10^{-9} s. Slow dynamics is characteristic of macromolecules systems and NMR relaxometry is a highly advantageous method for studying such motional processes [17,47,48]. Fig. 8 shows an example of ^1H spin-lattice relaxation data collected for gelated BSA.

The relaxation data exhibit frequency specific relaxation maxima often referred to as quadrupole peaks. They represent Quadrupole Relaxation Enhancement (QRE) effects [49–60], as indicated in Fig. 8. Before explaining the mechanism of the QRE effects we should first focus on the origin of the ^1H spin-lattice relaxation for proteins. The overall relaxation rate, $R_{1,H}(\omega_H)$, results from ^1H - ^1H and ^1H - ^{14}N magnetic dipole-dipole interactions:

$$R_{1,H}(\omega_H) = R_{1,H}^{HH}(\omega_H) + R_{1,H}^{HN}(\omega_H) \quad (21)$$

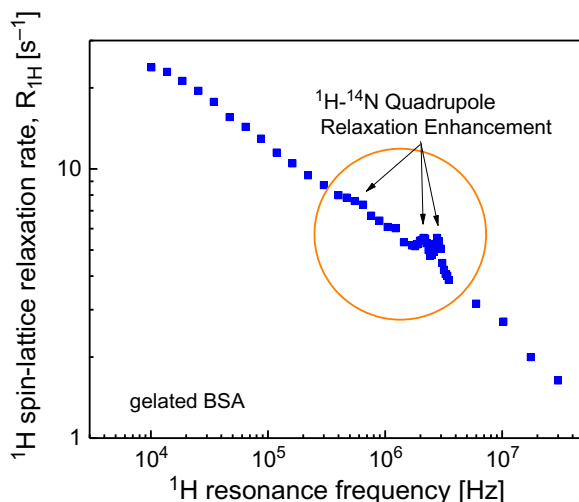


Fig. 8 ^1H spin-lattice relaxation rates for gelated BSA at 293 K.

where the $R_{1,H}^{HH}(\omega_H)$ and $R_{1,H}^{HN}(\omega_H)$ relaxation terms correspond to the two contributions, respectively. The $R_{1,H}^{HH}(\omega_H)$ relaxation rate can be expressed as:

$$R_{1,H}^{HH}(\omega_H) = C_{DD}^{HH}[J(\omega_H) + 4J(2\omega_H)] \quad (22)$$

Eq. (22) is a generalization of Eq. (21): $J_{rot}(\omega)$ has been replaced by $J(\omega)$, while the pre-factor $2I(I+1)\left(\frac{\mu_0 \gamma_I^2 \hbar}{4\pi r^3}\right)^2$ has been replaced by C_{DD}^{HH} , referred to as a dipolar relaxation constant. The first replacement reflects the fact that Eq. 22 is not limited to rotational dynamics and the spectral density can take any other form, while the second one account for the fact that in macromolecules there is many ^1H nuclei involved in a network of mutual dipole-dipole interactions; C_{DD}^{HH} can be treated as a quantity reflecting an effective amplitude of these interactions. The expression can be further generalized accounting for different ^1H - ^1H dipole-dipole interactions modulated by different dynamical processes:

$$R_{1,H}^{HH}(\omega_H) = \sum_n C_{DD,n}^{HH}[J_n(\omega_H) + 4J_n(2\omega_H)] \quad (23)$$

The relaxation data for elastin can be reproduced in terms of three motional processes referred to as slow, intermediate and fast ones (not discussing at this stage their origin). Consequently, the expression for $R_{1,H}^{HH}(\omega_H)$ can explicitly be written as:

$$\begin{aligned} R_{1,H}^{HH}(\omega_H) = & C_{DD,s}^{HH} \left[\frac{\tau_s}{1 + (\omega_H \tau_s)^2} + \frac{4\tau_s}{1 + (2\omega_H \tau_s)^2} \right] \\ & + C_{DD,i}^{HH} \left[\frac{\tau_i}{1 + (\omega_H \tau_i)^2} + \frac{4\tau_i}{1 + (2\omega_H \tau_i)^2} \right] \\ & + C_{DD,f}^{HH} \left[\frac{\tau_f}{1 + (\omega_H \tau_f)^2} + \frac{4\tau_f}{1 + (2\omega_H \tau_f)^2} \right] + A \end{aligned} \quad (24)$$

where the correlation times τ_s , τ_i and τ_f characterize the slow, intermediate and fast dynamics, while $C_{DD,s}^{HH}$, $C_{DD,i}^{HH}$ and $C_{DD,f}^{HH}$ are the corresponding dipolar relaxation constants. The frequency independent term, A , corresponds to a relaxation contributions from a pool of ^1H nuclei for which the ^1H - ^1H dipole-dipole coupling fluctuate very fast (faster than on the time scale of 10^{-9} s)—one can think here about the methyl group rotation. For so fast dynamics one gets $\omega_H \tau_f \ll 1$ (so called extreme narrowing case [4]) and therefore the spectral densities become independent of frequency. One

should be careful performing this type of parametrisation because using a high number of parameters one can, in principle, reproduce every set of relaxation data. The number of parameters must be limited (involving more than three relaxation terms in Eq. (24) would be an over-parametrization) and the parameters must be physically sound.

The quadrupole peaks are associated with the $R_{1,H}^{HN}(\omega_H)$ relaxation contribution. In analogy to Eq. (2) the close form expression for a relaxation rate $R_{1,I}^{IS}(\omega_I)$ resulting from a dipole-dipole coupling between spin $I = \frac{1}{2}$ and a spin S has the form [3–7,61–63]:

$$R_{1,I}^{IS}(\omega_I) = \frac{1}{2} \left(\frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \right)^2 S(S+1) [J(\omega_I - \omega_S) + 3J(\omega_I) + 6J(\omega_I + \omega_S)] \quad (25)$$

where ω_S denotes the S spin resonance frequency; $S=1$ for ^{14}N . The crucial point is, however, that independently of the form of the spectral density, $J(\omega)$, Eq. (25) will never create a frequency specific relaxation enhancement. The reason for that is that Eq. (25) is valid only as long as the Zeeman interaction of the S spin is the only source of its energy level structure (in other words, the S spin exhibits a Zeeman energy level structure). However, nuclei of $S \geq 1$ placed in an electric field gradient exhibit a quadrupole interaction. The quadrupole coupling contributes (dominates) the energy level structure of the S spin, provided the orientation of the principal axis system of the electric field gradient (P) is fixed (fluctuates very slowly) with respect to the direction of the external magnetic field, $\vec{B}_0$. The quadrupole Hamiltonian, $H_Q(S)$, takes in the (P) frame the form [6,7,64,65]:

$$H_Q(S) = \frac{1}{2} \sqrt{\frac{3}{2}} \frac{a_Q}{S(2S-1)} \left[T_0^2(S) + \frac{\eta}{\sqrt{6}} (T_{-2}^2(S) + T_2^2(S)) \right] \quad (26)$$

where the quadrupole coupling constant is defined as: $a_Q = \frac{e^2 q Q}{h}$ (Q denotes the quadrupole moment of the nucleus, while q is the zz component of the electric field gradient tensor at the position of the S spin nucleus), while the asymmetry parameter, η , is given as: $\eta = \frac{V_{xx} - V_{yy}}{V_{zz}}$ (V_{xx} , V_{yy} and V_{zz} are Cartesian components of the electric field gradient tensor). The spin tensor components are defined as: $T_0^2(S) = \frac{1}{\sqrt{6}} [3S_z^2 - S(S+1)]$, $T_{\pm 1}^2(S) = \mp \frac{1}{2} (S_z S_{\pm} + S_{\pm} S_z)$, $T_{\pm 2}^2(S) = \frac{1}{2} S_{\pm} S_{\pm}$. When the quadrupole interaction dominates the Zeeman coupling of the S spin, one can provide analytical expressions for the transition frequencies (energies) for $S=1$ between its

energy levels [49–55,58–60]: $\nu_- = \frac{3}{4}a_Q(1 - \frac{\eta}{3})$, $\nu_+ = \frac{3}{4}a_Q(1 + \frac{\eta}{3})$ and $\nu_0 = \nu_+ - \nu_- = \frac{1}{2}\eta a_Q$. At some magnetic fields the I spin (^1H in this case) resonance frequency matches one of these transition frequencies. The I and S spin transitions between their energy levels are coupled by the mutual $I-S$ dipole-dipole interactions. When the transition frequencies (energies) much, the S spin relaxation (transitions) enhances the relaxation (transitions) of the I spin. This effect is sometimes called cross-relaxation, although this term should rather be used in a different context. When the condition $\omega_Q\tau_Q \gg 1$ is fulfilled (the product should reach the value of at least 20), where $\omega_Q = 2\pi a_Q$ and τ_Q denotes the correlation time describing the fluctuations of the electric field gradient tensor, the $R_{1,H}^{HN}(\omega_H)$ relaxation rate (for $S=1$) can be expressed as [66]:

$$R_{1,H}^{HN}(\omega_H) = \frac{2}{3} \left(\frac{\mu_0 \gamma_H \gamma_N \hbar}{4\pi r^3} \right)^2 \left[\begin{aligned} & \left(\frac{1}{3} + \sin^2\Theta \cos^2\Phi \right) \left(\frac{\tau_Q}{1 + (\omega_H - \omega_-)^2 \tau_Q^2} + \frac{\tau_Q}{1 + (\omega_H + \omega_-)^2 \tau_Q^2} \right) + \\ & \left(\frac{1}{3} + \sin^2\Theta \sin^2\Phi \right) \left(\frac{\tau_Q}{1 + (\omega_H - \omega_+)^2 \tau_Q^2} + \frac{\tau_Q}{1 + (\omega_H + \omega_+)^2 \tau_Q^2} \right) + \\ & \left(\frac{1}{3} + \cos^2\Theta \right) \left(\frac{\tau_Q}{1 + (\omega_H - \omega_0)^2 \tau_Q^2} + \frac{\tau_Q}{1 + (\omega_H + \omega_0)^2 \tau_Q^2} \right) \end{aligned} \right] \quad (27)$$

where $\omega_{\pm} = 2\pi\nu_{\pm}$, $\omega_0 = 2\pi\nu_0$. The angles Θ and Φ describe the orientation of the $I-S$ dipole-dipole coupling in the principal axis system of the electric field gradient tensor.

The model represented by Eqs. (21), (24), and (27) has been applied to several solid state proteins in Ref. [67]. Quadrupole peaks are observed not only for biomolecules; this effect has also been reported for various kinds of solids [51–57,68–70]. As far as biological systems are considered QRE has been observed for tissues [60] and food products. Further examples of QRE effects are shown in Fig. 9 and Fig. 10.

The positions of the quadrupole peaks are determined by the quadrupole coupling of the S spin nucleus. This implies that the QRE effects allow to probe changes in the electronic and molecular surrounding of the nucleus leading to even subtle changes in the electric field gradient tensor. Such information is unique and not available by other methods.

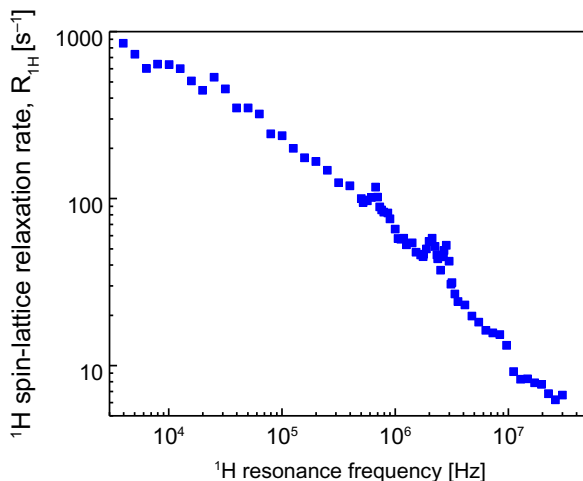


Fig. 9 ^1H spin-lattice relaxation rate for hydrated lysozyme (50%) at 293 K.

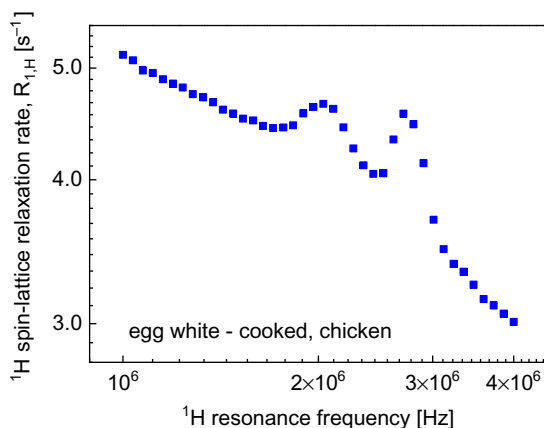


Fig. 10 ^1H spin-lattice relaxation rate for chicken egg white; well visible quadrupole peaks.

Before discussing further aspects of QRE effects one has to distinguish between relaxation enhancement and polarization transfer. The key factor is the time scale of the fluctuations of the $I-S$ dipole-dipole coupling. As already explained, to talk about a relaxation process the coupling must stochastically fluctuate in time. When the condition $\omega_{DD}\tau_c \ll 1$ is fulfilled (ω_{DD} denotes the amplitude of the dipole-dipole coupling in angular frequency units), one can describe the relaxation process in terms of the Redfield relaxation theory [3–6,27] (based on the perturbation solution of the Liouville equation). The condition implies that the fluctuations of

the dipole–dipole interaction causing the relaxation process is much faster than the relaxation itself. When this condition is fulfilled, one can talk about QRE. The situation can, however, be different. First, the dynamics can be so slow (for instance, for some solids at low temperatures) that the condition is not fulfilled. Second, the dynamics can be restricted. This can lead to a non-zero average of the dipole–dipole interactions (even when the dynamics is fast) referred to as a residual dipolar coupling. In both situations there is a time independent $I-S$ dipole–dipole coupling. In such a case, when the resonance frequency of the I spin nucleus matches one transitions frequencies of the S spin nucleus, the I spin magnetization can be ‘taken over’ by the quadrupole nucleus. This manifests itself as a frequency specific decrease (decay) of the I spin magnetization and can be (erroneously) attributed to QRE effects although there is no relaxation involved in this phenomenon. When the mutual, $I-S$, spin coupling is time independent, one should talk about a transfer of polarization (magnetization) [70–72]. The polarization transfer effect can be modelled by solving the time-independent Liouville equation. The difference between the QRE and polarization transfer effects is illustrated in Fig. 11 [72].

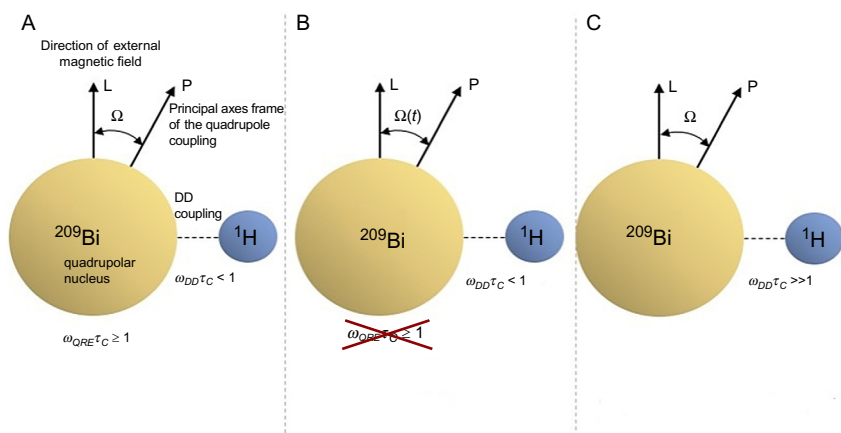


Fig. 11 Schematic representation of QRE and polarization transfer effects (^{209}Bi has been used as an example of nuclei possessing quadrupole moments). For the cases shown in (A) and (B) the ^1H -X (X denotes a quadrupole nucleus) dipole-dipole coupling acts as a relaxation mechanism, but only in the first case (A) quadrupole peaks can be observed. In the last case (C) the ^1H -X dipole-dipole coupling serves as a polarization transfer pathway. Reprinted from Ref. D. Kruk, E. Masiewicz, E. Umut, M. Schlögl, R. Fischer, H. Scharfetter, *Quadrupole relaxation enhancement and polarisation transfer in DMSO solution of $[\text{Bi}(\text{NO}_3)_3(\text{H}_2\text{O})_3]^*18\text{-crown-6}$ in solid state*, *Mol. Phys.* 117 (2019) 944–951, with permission.

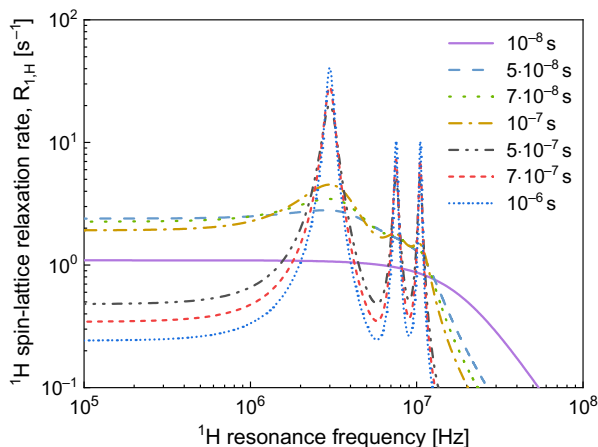


Fig. 12 QRE effects associated with ^1H - ^{14}N dipole-dipole interactions, depending on the correlation time, τ_Q .

Fig. 12 shows the progressive increasing of the amplitudes of the quadrupole peaks with increasing of the correlation time, τ_Q , describing the modulations of the ^1H - ^{14}N dipole-dipole coupling. The simulations have been performed in terms of a general (valid for arbitrary motional conditions) model of QRE [51,52,55,67,73] based on the stochastic Liouville Equation.

The quantum-mechanical picture of the both QRE and polarization transfer is, in fact, much more complex. There are several reasons for that; let us mention some of them. For instance, the fluctuations of the $I-S$ dipole-dipole coupling originates not only from the dynamics of the molecule carrying the I and S spins, but also from the relaxation of the quadrupole nucleus. In the simplest case, one can assume that S spin relaxation is caused by a mechanism which is independent of the presence of the neighbouring I spin nucleus. One can attribute the S spin relaxation to local fluctuations of the electric field gradient tensor. Assuming that the relaxation is independent of the magnetic field, one can set: $\frac{1}{\tau_Q} = \frac{1}{\tau_c} + R_Q$ (typically $\tau_c = \tau_{rot}$) where R_Q denotes the quadrupole spin relaxation. When $R_Q \ll \frac{1}{\tau_c}$, the quadrupole relaxation contribution to the fluctuations of the $I-S$ dipole-dipole interaction can be neglected. The assumption that the quadrupole spin relaxation is independent of the magnetic field can be an oversimplification. Moreover, one should be aware that the different transitions of the S spin can be associated with considerably different quadrupole relaxation rates, R_Q [74]. To make the scenario even more complex

one should note that the I and S spins can belong to different molecule and, in consequence, the spectral densities of Eq. (27) have to be replaced by expressions analogous to Eq. (14). Eventually, the assumption of the hierarchy of events: the S spin relaxation is independent of the presence of the I spin, but it influences the I spin relaxation, might not hold—the efficiency of the fluctuations of the electric field gradient tensor as the S spin relaxation does not necessarily exceeds the $I-S$ dipole-dipole relaxation contribution to the S spin relaxation [74].



4. Examples of QRE and polarization transfer

Above we have presented examples of QRE effects for solid and partially hydrated proteins. The ^1H spin-lattice relaxation enhancement is associated with ^{14}N nuclei of $S=1$. It has been pointed out that to observe QRE effects the system must undergo a relatively slow dynamics. This is why one does not see QRE effects for proteins in solution (Fig. 13)—the rotational dynamics of the protein molecules is too fast to fulfil the condition $\omega_Q\tau_Q \gg 1$. The tumbling of the protein molecules becomes faster for lower concentrations of the protein.

An analogous situation takes place for QRE effects associated with nuclei of high spin quantum numbers, for instance, ^{209}Bi of $S = \frac{9}{2}$. Fig. 14 shows ^1H spin-lattice relaxation data for pepto-bismol (a medication used to treat gastrointestinal discomforts); this is a colloidal substance obtained by

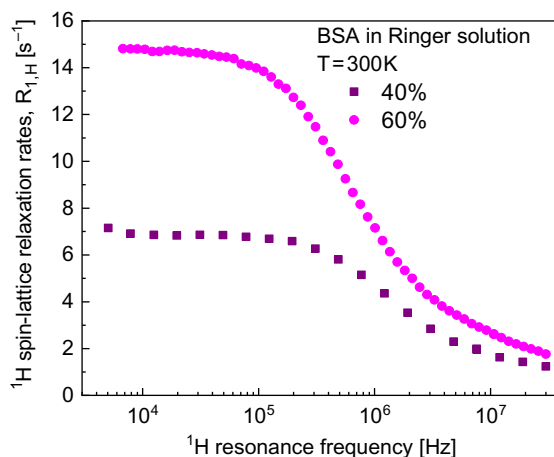


Fig. 13 ^1H spin-lattice relaxation data for BSA dissolved in Ringer solution at room temperature; the protein concentration yields 40% and 60%.

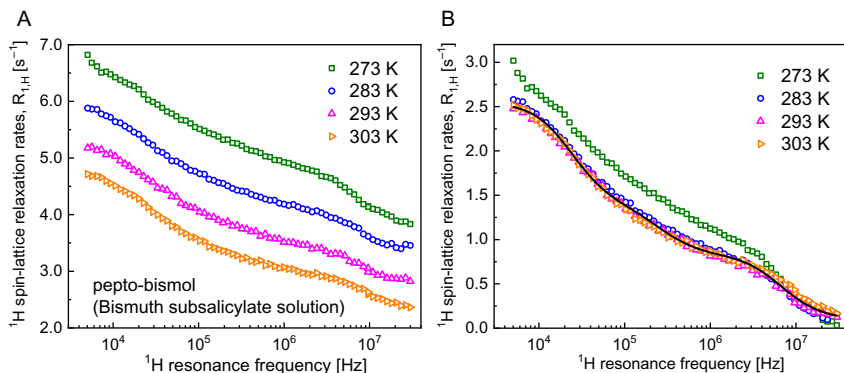


Fig. 14 ^1H spin-lattice relaxation rates for pepto-bismol (A), the same data normalized as described in the text (B); solid line—fit in terms of Eq. (24). The obtained parameters are given in the text.

hydrolysis of bismuth salicylate. The relaxation dispersion curves for 283, 293 and 303 K almost coincide after subtracting a frequency independent term, A , that yields: 3.3 s^{-1} (283 K), 2.7 s^{-1} (293 K) and 2.2 s^{-1} (303 K). The results for 273 K are different ($A = 3.0\text{ s}^{-1}$ in this case). The changes in the shapes of the relaxation profiles can be attributed to different water dynamics below 273 K and above 273 K. The data have been fitted in terms of Eq. (24). The three dynamical process, revealed in this way for 283, 293 and 303 K, are (approximately) characterized by the parameters: $C_{DD,s}^{HH} = 6.3 \times 10^4\text{ Hz}^2$, $\tau_s = 3.7 \times 10^{-6}\text{ s}$; $C_{DD,i}^{HH} = 3.3 \times 10^5\text{ Hz}^2$, $\tau_i = 3.4 \times 10^{-7}\text{ s}$; $C_{DD,f}^{HH} = 1.1 \times 10^7\text{ Hz}^2$, $\tau_f = 1.3 \times 10^{-8}\text{ s}$.

QRE effects are, however, clearly seen for solid bismuth subsalicylate, as shown in Fig. 15. They manifest themselves as a broad relaxation maximum in the frequency range of about 20–100 MHz. For high spin quantum numbers one can expect several QRE lines (there are already three for $S = 1$). We shall discuss this subject later. It is also worth to notice that QRE is not observed for frozen pepto-bismol although in this case one also can expect slow fluctuations of ^1H – ^{209}Bi dipole-dipole interactions.

They can be two reasons for the absence of quadrupole peaks for frozen pepto-bismol. The first one is the structural heterogeneity of the system. Frozen pepto-bismol is amorphous that implies different values of electric field gradient tensor at the ^{209}Bi positions. In consequence, the quadrupole peaks have low amplitudes and are spread over a broad frequency range. Moreover (and this is the second reason), due to their low amplitudes they

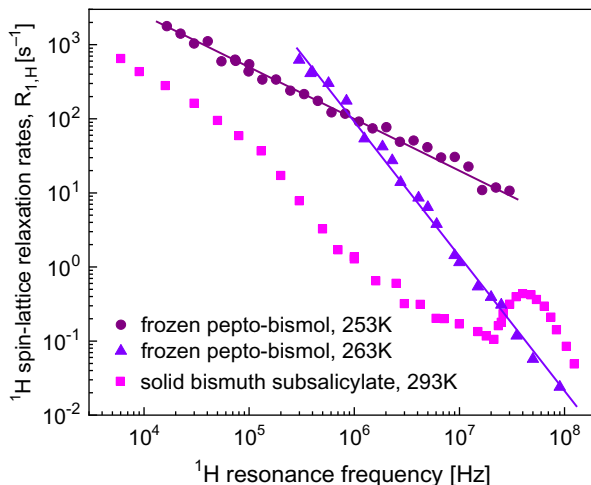


Fig. 15 ^1H spin-lattice relaxation rates for solid bismuth subsalicylate (295 K) and frozen pepto-bismol (263 and 253 K).

can be masked by the ^1H - ^1H relaxation contribution. It is also worth to point out the linear shape of the ^1H relaxation dispersion profiles for frozen pepto-bismol. Such a shape follows a power law dependence: $R_{1,H}(\omega_H) \propto A\omega^{-\alpha}$ [48,75,76] characteristic of, so called, sub-diffusive processes, for which the mean square displacement does not change linearly with time. The concept of a power-law has been exploited to interpret ^1H spin-lattice relaxation data for polypeptides and proteins [77–82]. The dynamics has been attributed to vibrations (turbulences) propagating along the polypeptide backbones.

An interesting example of ^{19}F spin-lattice relaxation enhancement originating from ^{19}F - ^{209}Bi dipole-dipole interactions has been observed for BiF_3 as shown in Fig. 16. Bismuthtrifluoride is an ionic solid with orthorhombic crystal structure. It has common β - BiF_3 phase having a YF_3 type structure, composing of a nine-coordinate ^{209}Bi - $^{19}\text{F}_9$ tricapped trigonal prism with eight approximately equal ^{209}Bi - ^{19}F bond lengths (between 2.217 and 2.502 Å) and one longer ^{209}Bi - ^{19}F distance (3.1 Å); the unit cell volume of BiF_3 is very high (222.8 Å³). The ^{19}F - ^{19}F distances vary from 2.693 Å to 3.429 Å [83].

For BiF_3 the QRE effect takes the form of a very broad peak (much broader than even for bismuth subsalicylate) that very considerably changes its shape and position with temperature. One should stress at this stage that

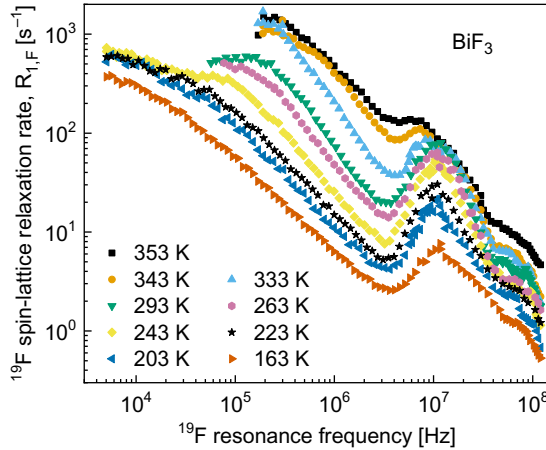


Fig. 16 ^{19}F spin-lattice relaxation rates for BiF_3 .

BiF_3 does not undergo any phase transition in the whole temperature range. As far as the temperature evolution of the QRE effects is concerned, one can expect that with increasing temperature the effects become progressively less pronounced as the dynamics becomes faster (provided there is no phase transition changing this trend). The experimental data for BiF_3 confirm this overall tendency. One can, however, also clearly seen from Fig. 16 that the relaxation contribution originating from ^{19}F - ^{19}F dipole-dipole interactions increases with temperature masking the ^{19}F - ^{209}Bi relaxation contribution.

As already pointed out, for large S values one can expect many quadrupole peaks. This expectation has not been satisfied neither for bismuth subsalicylate nor BiF_3 . One can explain that by fast quadrupole relaxation leading to a significant broadening of the individual lines that, in consequence, overlap. A perfect example of well resolved QRE effects can be seen for tris(2-methoxyphenyl)bismuthane in Fig. 17 [56]. The figure has been constructed as follows. In the first step, the total Hamiltonian of the S spin, $H_{\text{tot}}(S)$ has been set up. The Hamiltonian includes the Zeeman coupling of the S spin represented by the Hamiltonian $H_Z(S)$, and its quadrupole coupling expressed in the laboratory (L) frame, $H_Q^{(L)}(S)$ [6,7,52,56,64,65,84]:

$$H_Q^{(L)}(S) = \frac{1}{2} \sqrt{\frac{3}{2}} \frac{a_Q}{S(2S-1)} \sum_{n=-2}^2 (-1)^n V_{-n}^{2(L)}(\Omega_{PL}) T_n^2(S) \quad (27)$$

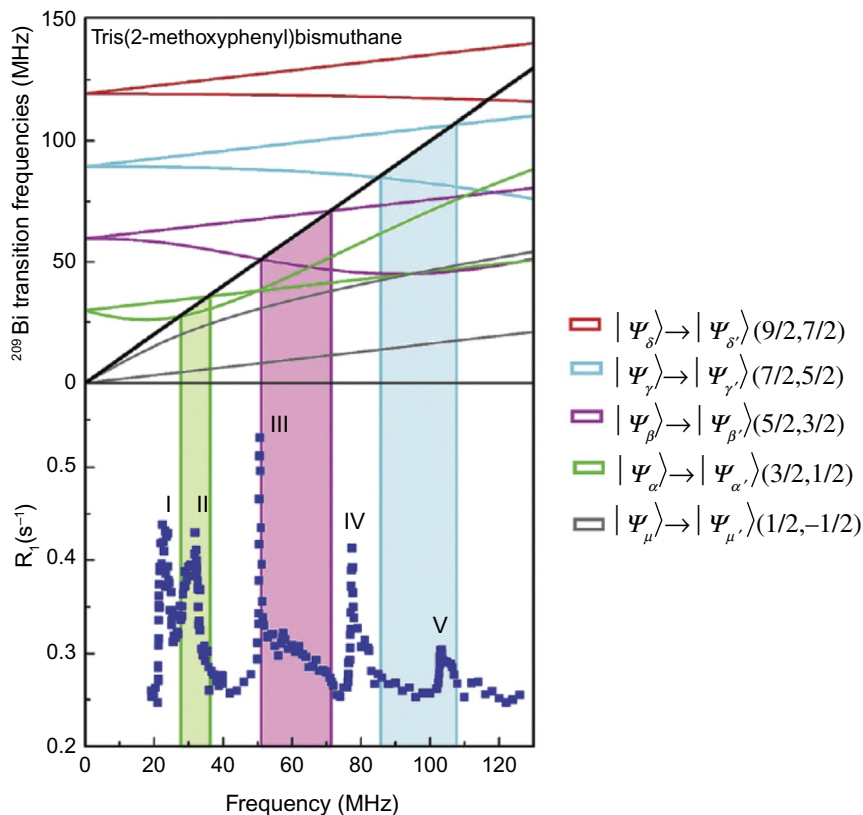


Fig. 17 QRE effects for tris(2-methoxyphenyl)bismuthane originating from single-quantum transitions. Reprinted from Ref. D. Kruk, E. Umut, E. Masiewicz, C. Sampl, R. Fischer, S. Spirk, C. Goesweiner, H. Scharfetter, *209Bi quadrupole relaxation enhancement in solids as a step towards new contrast mechanisms in magnetic resonance imaging*, *Phys. Chem. Chem. Phys.* 20 (2018) 12710–12718, with permission.

where $V_{-n}^{2(L)} = D_{0,-n}^2(\Omega_{PL}) + \frac{\eta}{\sqrt{6}} [D_{-2,2}^2(\Omega_{PL}) + D_{2,2}^2(\Omega_{PL})]$. The laboratory form of the quadrupole Hamiltonian has been obtained by using the transformation of Eq. (5), where the angle Ω_{PL} describes the orientation of the principal axis system of the electric field gradient tensor with respect to the direction of the external magnetic field. In the next step it is necessary to calculate the values of the available energy levels (eigenvalues) and the corresponding eigenvectors. For this purpose one has to set up the matrix representation of the $H_{tot}(S)$ Hamiltonian in the $\{|m_S\rangle\}$ basis. The size of the Hamiltonian matrix is $(2S+1)$; the index (L) in $V_{-n}^{2(L)}$ has been omitted for simplicity, analogously it has been set: $\Omega_{PL} = \Omega$:

$$H_Q^{(L)}(S) = \frac{1}{72} \sqrt{\frac{3}{2}} a_Q \times$$

$$\times \begin{bmatrix} Q & |\frac{9}{2}\rangle & |\frac{7}{2}\rangle & |\frac{5}{2}\rangle & |\frac{3}{2}\rangle & |\frac{1}{2}\rangle & |-\frac{1}{2}\rangle & |-\frac{3}{2}\rangle & |-\frac{5}{2}\rangle & |-\frac{7}{2}\rangle & |-\frac{9}{2}\rangle \\ \langle 9/2| & \frac{9}{2}\omega_S + 6\sqrt{6}V_0^2(\Omega) & 12V_{-1}^2(\Omega) & 6V_{-2}^2(\Omega) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \langle 7/2| & -12V_1^2(\Omega) & \frac{7}{2}\omega_S + 2\sqrt{6}V_0^2(\Omega) & 12V_{-1}^2(\Omega) & 2\sqrt{21}V_{-2}^2(\Omega) & 0 & 0 & 0 & 0 & 0 & 0 \\ \langle 5/2| & 6V_2^2(\Omega) & -12V_1^2(\Omega) & \frac{5}{2}\omega_S - \sqrt{6}V_0^2(\Omega) & 2\sqrt{21}V_{-1}^2(\Omega) & 3\sqrt{14}V_{-2}^2(\Omega) & 0 & 0 & 0 & 0 & 0 \\ \langle 3/2| & 0 & 2\sqrt{21}V_2^2(\Omega) & -2\sqrt{21}V_1^2(\Omega) & \frac{3}{2}\omega_S - 3\sqrt{6}V_0^2(\Omega) & 2\sqrt{6}V_{-1}^2(\Omega) & 5\sqrt{6}V_{-2}^2(\Omega) & 0 & 0 & 0 & 0 \\ \langle 1/2| & 0 & 0 & 3\sqrt{14}V_2^2(\Omega) & -2\sqrt{6}V_1^2(\Omega) & \frac{1}{2}\omega_S - 4\sqrt{6}V_0^2(\Omega) & 0 & 5\sqrt{6}V_{-2}^2(\Omega) & 0 & 0 & 0 \\ \langle -1/2| & 0 & 0 & 0 & 5\sqrt{6}V_2^2(\Omega) & 0 & -\frac{1}{2}\omega_S - 4\sqrt{6}V_0^2(\Omega) & -2\sqrt{6}V_{-1}^2(\Omega) & 3\sqrt{14}V_{-2}^2(\Omega) & 0 & 0 \\ \langle -3/2| & 0 & 0 & 0 & 0 & 5\sqrt{6}V_2^2(\Omega) & 2\sqrt{6}V_1^2(\Omega) & -\frac{3}{2}\omega_S - 3\sqrt{6}V_0^2(\Omega) & -2\sqrt{21}V_{-1}^2(\Omega) & 2\sqrt{21}V_{-2}^2(\Omega) & 0 \\ \langle -5/2| & 0 & 0 & 0 & 0 & 0 & 3\sqrt{14}V_2^2(\Omega) & 2\sqrt{21}V_1^2(\Omega) & -\frac{5}{2}\omega_S - \sqrt{6}V_0^2(\Omega) & -12V_{-1}^2(\Omega) & 6V_{-2}^2(\Omega) \\ \langle -7/2| & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{21}V_2^2(\Omega) & 12V_1^2(\Omega) & -\frac{7}{2}\omega_S + 2\sqrt{6}V_0^2(\Omega) & -12V_{-1}^2(\Omega) \\ \langle -9/2| & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6V_2^2(\Omega) & 12V_1^2(\Omega) & -\frac{9}{2}\omega_S + 6\sqrt{6}V_0^2(\Omega) \end{bmatrix}$$

(28)

For comparison, the matrix representation of $H_{\text{tot}}(S)$ for $S=1$ is given by Eq. (29):

$$[H_{\text{tot}}(S)] = \begin{bmatrix} -\omega_S + \frac{1}{4}a_Q V_0^{2(L)}(\Omega_{PL}) & -\frac{\sqrt{3}}{4}a_Q V_1^{2(L)}(\Omega_{PL}) & \frac{1}{2}\sqrt{\frac{3}{2}}a_Q V_2^{2(L)}(\Omega_{PL}) \\ \frac{\sqrt{3}}{4}a_Q V_{-1}^{2(L)}(\Omega_{PL}) & -\frac{1}{2}a_Q V_0^{2(L)}(\Omega_{PL}) & -\frac{\sqrt{3}}{4}a_Q V_1^{2(L)}(\Omega_{PL}) \\ \frac{1}{2}\sqrt{\frac{3}{2}}a_Q V_{-2}^{2(L)}(\Omega_{PL}) & \frac{\sqrt{3}}{4}a_Q V_{-1}^{2(L)}(\Omega_{PL}) & \omega_S + \frac{1}{4}a_Q V_0^{2(L)}(\Omega_{PL}) \end{bmatrix} \quad (29)$$

When the Zeeman contribution to the $H_{\text{tot}}(S)$ is neglected in Eq. (29), one obtains for $S=1$ the following energy levels (eigenvalues of the Hamiltonian matrix): $E_1 = \frac{1}{4}a_Q(1 - \eta)$, $E_2 = -\frac{1}{2}a_Q$, $E_3 = \frac{1}{4}a_Q(1 + \eta)$, as a result of its diagonalization. In this case one can hardly discuss the relative orientation of the principal axis system of the electric field gradient tensor with respect to the direction of the external magnetic field, so one can set for practical purposes $\Omega_{PL}=0$. In this way one gets the matrix representation of the Hamiltonian of Eq. (26).

Coming back to the case of $S = \frac{9}{2}$, for ^{209}Bi one should not neglect the Zeeman interaction because the relatively high gyromagnetic factor of ^{209}Bi combined with the high spin quantum number that makes the Zeeman contribution even more important (see the diagonal elements of the matrix of Eq. (28)). In consequence, the energy levels (resulted from diagonalization of the matrix of Eq. (28)) become dependent on the orientation represented by the Ω_{PL} angle. The same applies to the transition energies (expressed in the frequency units). The upper part of Fig. 17 shows transition frequencies (energies) between some of the available quantum states for all possible values of Ω_{PL} . In the high field range, when the Zeeman coupling dominates the quadrupole interaction, the complex eigenvectors (given as linear combinations of the $\{|m_S\rangle\}$ vectors converge to single $|m_S\rangle$ vectors. The transition frequencies drawn in Fig. 17 correspond to the pairs of energy levels for which the magnetic quantum numbers of the corresponding high field states differ by one; such transitions are referred to as single quantum coherences. The single-quantum transition frequencies are compared with the ^1H resonance frequency and the areas in which these two frequencies match are marked. Then the areas are projected onto the experimental ^1H spin-lattice relaxation data (Fig. 17, bottom) to explain the physical origin of the quadrupole peaks. QRE effects can also stem from double-quantum transitions of

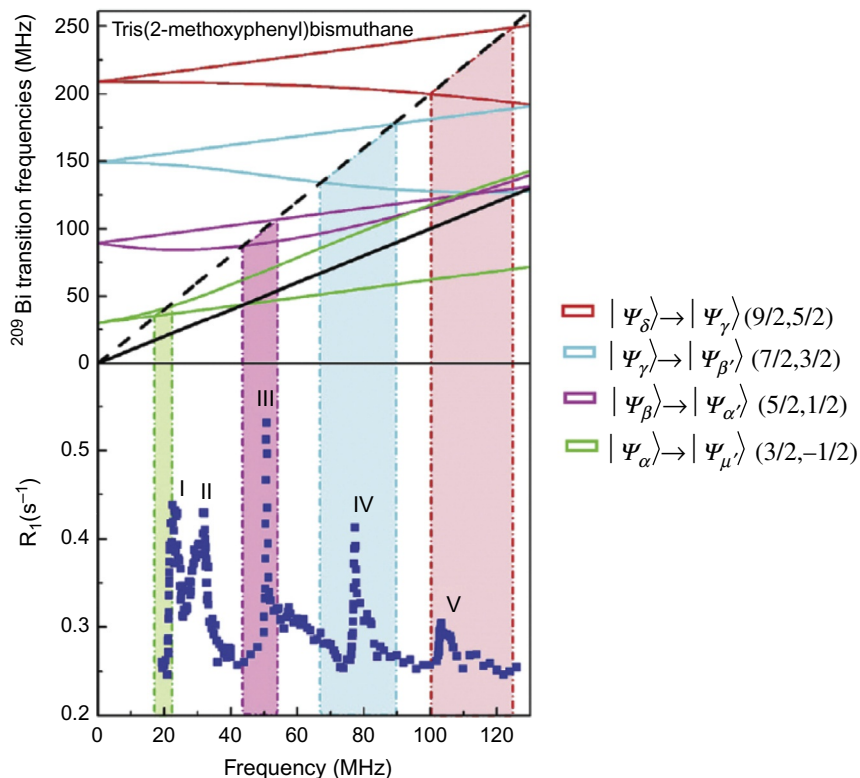


Fig. 18 QRE effects for tris(2-methoxyphenyl)bismuthane originating from double-quantum ^{209}Bi transitions combined with single-quantum and double-quantum ^1H transitions. Reprinted from Ref. D. Kruk, E. Umut, E. Masiewicz, C. Sampl, R. Fischer, S. Spirk, C. Goesweiner, H. Scharfetter, *^{209}Bi quadrupole relaxation enhancement in solids as a step towards new contrast mechanisms in magnetic resonance imaging*, *Phys. Chem. Chem. Phys.* 20 (2018) 12710–12718, with permission.

the quadrupole nucleus (of spin S) and/or double-quantum transitions of the spin $I = \frac{1}{2}$ nuclei. In the last case at least two mutually coupled (by a dipole-dipole interaction) I spin nuclei have to participate in this process. Examples of QRE and polarization transfer effects including double-quantum transitions of coupled ^1H nuclei have been shown in Ref. [68,69]. Therefore, here the case of double-quantum ^{209}Bi transitions is illustrated in Fig. 18. The figure shows that some of the quadrupole peaks seen in Fig. 17 for triphenylbismuth can be explained in this way. The figure also includes the doubled ^1H resonance frequency, $2\omega_I$, (red line, top) corresponding to double-quantum transitions of a pair of coupled ^1H nuclei.



5. Numerous relaxation pathways

We have already discussed two situations including more than one relaxation pathways. In the first case it has been pointed out that one should be aware of the existence of intra-molecular and inter-molecular ^1H - ^1H dipole-dipole interactions for liquids, contributing to the overall relaxation. In the second case it has been shown that for systems including nuclei possessing quadrupole moments (referred to as ‘quadrupole nuclei’) one should consider besides the ^1H - ^1H (^{19}F - ^{19}F) relaxation contributions, also the dipole-dipole coupling with the quadrupole nuclei. The relaxation scenario becomes even more complex when the systems include more NMR active nuclei. An example of such systems is ionic liquids composed of ^1H contacting cations and ^{19}F containing anions. A schematic representation of the relaxation pathways is shown in Fig. 19.

The ^1H spin-lattice relaxation includes three contributions originating from the ^1H - ^1H dipole-dipole intra-cation interaction, the ^1H - ^1H interaction coupling and the ^1H - ^{19}F cation-anion interaction, respectively. The first coupling is modulated by the rotational dynamics of the cation, the second one by the relative, cation-cation translation diffusion, while the last one by the relative, cation-anion translation motion. This implies that the ^1H spin-lattice relaxation rate, $R_{1,H}(\omega_H)$, is given as [46,85]:

$$R_{1,H}(\omega_H) = R_{1,H}^{\text{intra,cation}}(\omega_H) + R_{1,H}^{\text{cation-cation}}(\omega_H) + R_{1,H}^{\text{cation-anion}}(\omega_H) \quad (30)$$

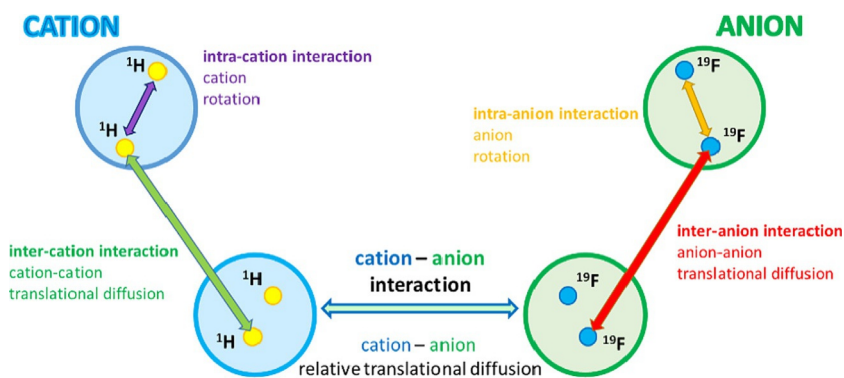


Fig. 19 Schematic representation of relaxation pathways in ionic liquids composed of ^1H containing cations and ^{19}F containing anions.

where:

$$\begin{aligned} R_{1,H}^{\text{intra,cation}}(\omega_H) &= C_{DD}^{\text{cation}} [J_{\text{rot,cation}}(\omega_H) + 4J_{\text{rot,cation}}(2\omega_H)] \\ &= C_{DD}^{\text{cation}} \left[\frac{\tau_{\text{rot,cation}}}{1 + (\omega_H \tau_{\text{rot,cation}})^2} + \frac{4\tau_{\text{rot,cation}}}{1 + (2\omega_H \tau_{\text{rot,cation}})^2} \right] \end{aligned} \quad (31a)$$

In Eq. (30) it has been assumed that the rotational dynamics of the cation is isotropic, with the correlation time $\tau_{\text{rot,cation}}$; C_{DD}^{cation} denotes the corresponding dipolar relaxation constant. The $R_{1,H}^{\text{cation-cation}}(\omega_H)$ relaxation term can be expressed as:

$$\begin{aligned} R_{1,H}^{\text{cation-cation}}(\omega_H) &= \frac{108}{5} \left(\frac{\mu_0}{4\pi} \gamma_H^2 \hbar \right)^2 \frac{N_H}{d_{\text{cation-cation}}^3} * \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \\ &\quad \left[\frac{u^2 \tau_{\text{trans,cation-cation}}}{u^4 + (\omega_H \tau_{\text{trans,cation-cation}})^2} + \frac{4u^2 \tau_{\text{trans,cation-cation}}}{u^4 + (2\omega_H \tau_{\text{trans,cation-cation}})^2} \right] du \end{aligned} \quad (31b)$$

where $\tau_{\text{trans,cation-cation}} = \frac{d_{\text{cation-cation}}^2}{D_{\text{trans,cation-cation}}}$. The parameters have the following meaning: $d_{\text{cation-cation}}$ denotes the distance of closest approach for a pair of the cations (it can be approximated by the diameter of the cation), $D_{\text{trans,cation-cation}}$ is the relative translation diffusion coefficient for the cations. Assuming that the translational movement is uncorrelated, one can set: $D_{\text{trans,cation-cation}} = 2D_{\text{trans,cation}}$ ($D_{\text{trans,cation}}$ denotes the self-diffusion coefficient of the cation). This relationship is, however, not obvious—if the ions form clusters, their relative diffusion coefficient can be much smaller than a sum of their self-diffusion coefficients). One can argue that ions of the same sign hardly form clusters, but such effects have been observed [86]. N_H is the number of ^1H nuclei per unit volume of the liquid. The expression for $R_{1,H}^{\text{cation-anion}}(\omega_H)$ is more complex. The reason for that is that as the ^1H and ^{19}F nuclei are not equivalent (their resonance frequencies are different), Eq. (2) does not apply anymore and it has to be replaced by a different form:

$$\begin{aligned} R_{1,H}^{\text{cation-anion}}(\omega_H) &= \frac{36}{5} \left(\frac{\mu_0}{4\pi} \gamma_H \gamma_F \hbar \right)^2 \frac{N_F}{d_{\text{cation-anion}}^3} \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \\ &\quad * \left[\frac{u^2 \tau_{\text{trans,cation-anion}}}{u^4 + ((\omega_H - \omega_F) \tau_{\text{trans,cation-anion}})^2} + \frac{3u^2 \tau_{\text{trans,cation-anion}}}{u^4 + (\omega_H \tau_{\text{trans,cation-anion}})^2} \right. \\ &\quad \left. + \frac{6u^2 \tau_{\text{trans,cation-anion}}}{u^4 + ((\omega_H + \omega_F) \tau_{\text{trans,cation-anion}})^2} \right] du \end{aligned} \quad (31c)$$

Analogously to Eq. (31b) one gets: $\tau_{\text{trans,cation-anion}} = \frac{d_{\text{cation-anion}}^2}{D_{\text{trans,cation-anion}}}$, where $d_{\text{cation-anion}}$ is the distance of closest approach of the ions of different signs (it can be approximated by a sum of the radii of the ions), while the relative, cation-anion translational diffusion coefficient fulfils the relationship: $D_{\text{trans,cation-anion}} \leq D_{\text{trans,cation}} + D_{\text{trans,anion}}$ ($D_{\text{trans,anion}}$ denotes the self-diffusion coefficient of the anion). The parameter N_F is the number of ^{19}F nuclei per unit volume of the liquid, while ω_F is the ^{19}F resonance frequency (γ_F is the ^{19}F gyromagnetic factor). On this basis one can straightforwardly formulate the expressions for the ^{19}F spin-lattice relaxation rate: $R_{1,F}(\omega_F)$:

$$R_{1,F}(\omega_F) = R_{1,F}^{\text{intra,anion}}(\omega_F) + R_{1,F}^{\text{anion-anion}}(\omega_F) + R_{1,F}^{\text{anion-cation}}(\omega_F) \quad (32)$$

The contributions to the $R_{1,F}(\omega_F)$ relaxation rate can be expressed analogously to Eqs. (31a), (31b), (31c) for the ^1H relaxation:

$$R_{1,F}^{\text{intra,anion}}(\omega_F) = C_{DD}^{\text{anion}} \left[\frac{\tau_{\text{rot,anion}}}{1 + (\omega_F \tau_{\text{rot,anion}})^2} + \frac{4\tau_{\text{rot,anion}}}{1 + (2\omega_F \tau_{\text{rot,anion}})^2} \right] \quad (33a)$$

where now $\tau_{\text{rot,anion}}$ denotes the rotational correlation time of the anion and C_{DD}^{anion} is the corresponding dipolar relaxation constant; then:

$$R_{1,F}^{\text{anion-anion}}(\omega_F) = \frac{108}{5} \left(\frac{\mu_0}{4\pi} \gamma_F^2 \hbar \right)^2 \frac{N_F}{d_{\text{anion-anion}}^3} * \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \left[\frac{u^2 \tau_{\text{trans,anion-anion}}}{u^4 + (\omega_F \tau_{\text{trans,anion-anion}})^2} + \frac{4u^2 \tau_{\text{trans,anion-anion}}}{u^4 + (2\omega_F \tau_{\text{trans,anion-anion}})^2} \right] du \quad (33b)$$

where $\tau_{\text{trans,anion-anion}} = \frac{d_{\text{anion-anion}}^2}{D_{\text{trans,anion-anion}}}$ and:

$$R_{1,F}^{\text{anion-cation}}(\omega_F) = \frac{36}{5} \left(\frac{\mu_0}{4\pi} \gamma_H \gamma_F \hbar \right)^2 \frac{N_H}{d_{\text{cation-anion}}^3} \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} * \left[\frac{u^2 \tau_{\text{trans,cation-anion}}}{u^4 + ((\omega_H - \omega_F) \tau_{\text{trans,cation-anion}})^2} + \frac{3u^2 \tau_{\text{trans,cation-anion}}}{u^4 + (\omega_F \tau_{\text{trans,cation-anion}})^2} + \frac{6u^2 \tau_{\text{trans,cation-anion}}}{u^4 + ((\omega_H + \omega_F) \tau_{\text{trans,cation-anion}})^2} \right] du \quad (33c)$$

Two conclusions should be drawn from this description. The first one is that one may not neglect the cation-anion relaxation contribution in the data analysis. In other words, one should not treat ^1H and ^{19}F relaxation data as a source of information on the cation and anion dynamics solely—the

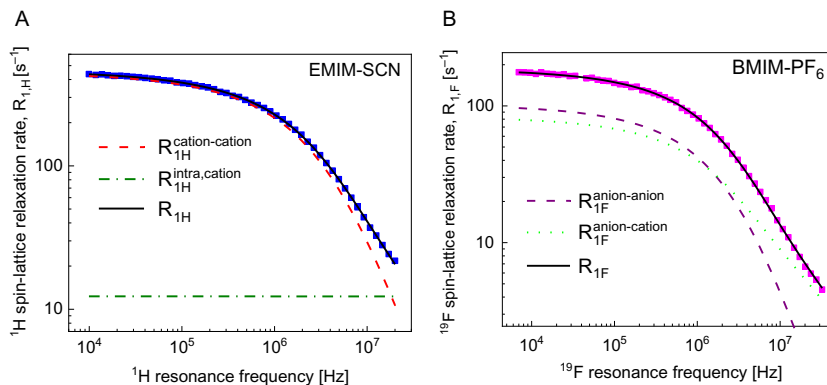


Fig. 20 ^1H spin-lattice relaxation data for EMIM-SCN at 263 K (A) and ^{19}F spin-lattice relaxation data for BMIM-PF $_6$ at 263 K (B); the relaxation dispersion profiles have been decomposed into the individual relaxation contributions.

relaxation processes are intertwined. The second conclusion is that the relaxation terms $R_{1,H}^{\text{cation-anion}}(\omega_H)$ and $R_{1,F}^{\text{anion-cation}}(\omega_F)$ can be much different, mostly due to different N_H and N_F numbers. To understand the importance of the cation-anion relaxation contribution one should compare Fig. 20A and B.

In Fig. 20A ^1H spin-lattice relaxation data for EMIM-SCN have been shown. As the anion does not include NMR active nuclei of a high gyromagnetic factor, the ^1H relaxation is, in a good approximation, caused solely by ^1H - ^1H dipole-dipole interactions with the predominant contribution associated with the cation-cation dipole-dipole coupling modulated by their relative diffusion. The rotational dynamics of EMIM cations has turned out to be too fast to lead to the $R_{1,H}^{\text{intra,cation}}(\omega_H)$ relaxation contribution being frequency dependent. Fig. 20B shows that the ^{19}F spin-lattice relaxation rates include a very significant contribution originating from the ^{19}F - ^1H anion-cation dipole-dipole coupling. The need of including all the relaxation channels should not be treated as an obstacle, but as a unique opportunity to probe the relative cation-anion translation motion.



6. Dynamics in confinement

Properties of liquids in confinement has attracted a lot of attention in recent years. The research is, in most cases, directed towards water (because of its relevance to biology) and polymers [17]. Physical and chemical properties of confined molecules are much different than in bulk.

Studies concerned with phase transition behaviour show that geometrical restrictions lead to a shift of glass transition temperature, melting and freezing points; for instance, water confined in nanotubes shows a liquid-like behaviour at temperatures as low as about 50 K [87]. Such studies have been accompanied by Molecular Dynamics simulations attempting to explain the observed phenomena on molecular level [87,88]. Dynamics of molecular systems (water, some other simple liquids and polymers) has also been investigated by NMR relaxometry. The studies have revealed peculiar dynamical mechanisms of molecular liquids in confinement, like for instance bulk mediated surface diffusion [89,90], proving the potential of NMR relaxometry as a very sensitive probe of motional mechanisms. For ionic liquids the influence of the confining walls on dynamics of ionic liquids is expected to be entirely different from that for molecular liquids. Moreover, as dynamics of the ions is affected by ionic interactions, one expects quite unusual dynamical effects. The expectations are confirmed by studies of the phase transition behaviour of ionic liquids in confinement which reveal that indeed the melting point, crystallization temperature and glass transition temperature significantly change in confinement [91–93].

Trajectories of molecules near the surface can be described as a succession of adsorption steps and flights in the ‘free’ fraction. One can distinguish here between weak adsorption and strong adsorption cases [90]. In the weak adsorption case molecules return to the ‘free’ fraction after a single adsorption act. In the strong adsorption case the molecules frequently return to the surface before they indeed escape to the ‘free’ fraction. During the sequence of ‘adsorption, desorption, movement near the surface, adsorption ...’ the molecules effectively perform 2D diffusion on the surface. Combining statistical description of Brownian motion near a surface [94] with spin relaxation theory one can access statistical properties of times and displacements between two surface hits—for instance, one can determine the average adsorption time and the minimal duration of the loop near the surface [85]. The subject of the dimensionality of the ionic (molecular) motion near surfaces raises controversies. Generally it is believed that the dynamics is of two-dimensional [75]. We shall come back to this problem later. Fig. 21 shows a comparison of ^1H spin-lattice relaxation dispersion profiles for EMIM-SCN in bulk and in a confinement created by SiO_2 matrix.

Fig. 21 shows that the relaxation dispersion profiles for bulk and confined liquid are much different in the whole frequency range indicating essential changes of the dynamical properties in confinement. For confined liquids one can think about two fractions of molecules: molecules adsorbed on

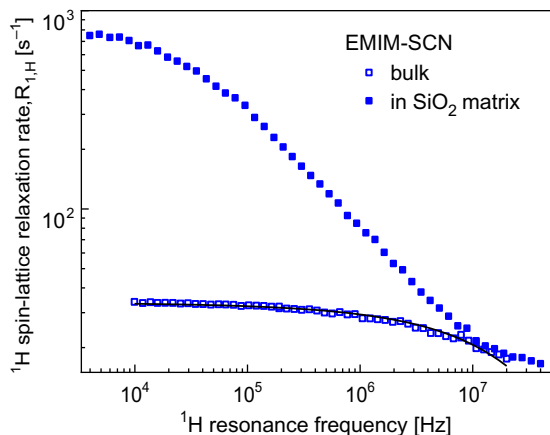


Fig. 21 ^1H spin-lattice relaxation dispersion data for EMIM-SCN in bulk and confined in SiO_2 matrix at 253 K.

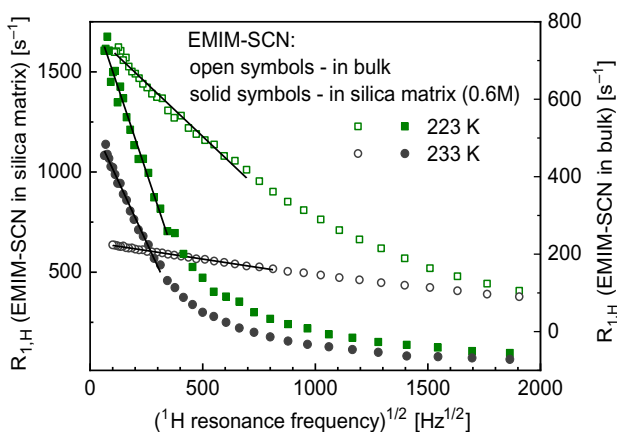


Fig. 22 ^1H spin-lattice relaxation rates for EMIM-SCN confined in SiO_2 matrix plotted vs a square root of the ^1H resonance frequency—ranges of linearity indicated by solid lines; the corresponding data for EMIM-SCN in bulk shown for comparison.

the surface (in this case oxygen of SiO_2 interacts with C-H bounds of EMIM cations) and a fraction of ‘free’ molecules. For EMIM-SCN in the confinement the translation diffusion of the ‘surface’ cations is by two orders of magnitude slower (the diffusion coefficient is of the order of $10^{-14} \text{ m}^2/\text{s}$) than for the cations in bulk ($10^{-12} \text{ m}^2/\text{s}$) [46]. Such pronounced effects are not observed for molecular liquids. To reveal the dimensionality of the diffusion of EMIM cations in the vicinity of the confining walls, in Fig. 22 the relaxation rates, $R_{1,H}$, obtained for EMIM-SCN in the confinement have been

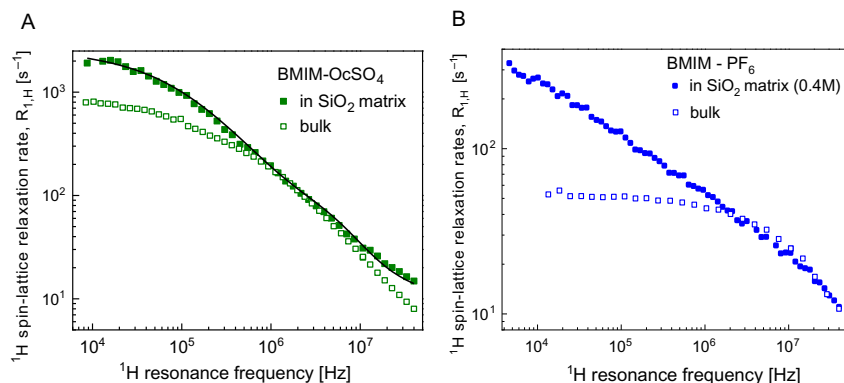


Fig. 23 ^1H spin-lattice relaxation data for (A) BMIM-OcSO₄ in bulk and confined in SiO₂ matrix at 263 K, (B) BMIM-PF₆ in bulk and confined in SiO₂ matrix at 288 K. *Republished with permission from D. Kruk, M. Wojciechowski, Sz. Brym, R.K. Singh, Dynamics of ionic liquids in bulk and in confinement by means of 1H NMR relaxometry-BMIM-OcSO₄ in an SiO₂ matrix as an example, Phys. Chem. Chem. Phys. 18 (2016) 23184–23194; permission conveyed through Copyright Clearance Center, Inc, Royal Society of Chemistry.*

plotted vs a square root of the ^1H resonance frequency (the corresponding results for EMIM-SCN in bulk, already shown in Fig. 5, have been added for comparison). The dependence is linear and this effect is, as already explained, a fingerprint of 3D (not 2D) translational diffusion.

Fig. 23 shows further examples of ^1H spin-lattice relaxation data for ionic liquids in SiO₂ confinement. In both cases the relaxation dispersion profiles for liquids in bulk and in the confinement are much different.

Nanofluids form a special class of confined systems. Recently nanofluids have gained attention for thermal conductivity enhancement and heat capacity effects caused by adding nanoobjects. The properties are highly dependent on the suspended material, which one can divide into three groups depending on its dimension: nanoparticles (0D), nanotubes (1D) and graphene (2D). In recent years investigations of ionic liquid based nanofluids containing nanoparticles have been carried out with the focus on their thermal properties, reporting enhancement of thermal conductivity; for instance, Al₂O₃ [95]. As far as heat capacity is concerned the first papers on this subject appeared in 2008 [96] for Al₂O₃ nanoparticles. Since this time some works have been done on heat capacity of other nanofluids; for instance, nanofluids composed of multiwalled carbon nanotubes in BMIM-BF₄ and BMIM-PF₆ have been studied by Ref. [97].

Graphene is a very new candidate for nanofluids [98,99]. As graphene is 2D object the heat transport properties are supposed to be different from

nanoparticles (0D) and carbon nanotubes (1D) and it is expected that such nanofluids show significant thermal conductivity enhancement (graphene is an excellent thermal conductor). Graphene has the largest surface area compared to nanotubes and nanoparticles, thus it has been expected that it considerably improves the effective thermal conductivity. The expectations have been confirmed for graphene-oxide ethylene glycol nanofluid [100] and graphene in water [98]. The addition of nanoobjects to fluids changes their dynamical behaviour. The outlined investigations focus on the application of ionic liquids as dispersion media for different kind of nanoobjects; however, there are only few studies attempting at answering questions regarding the molecular processes involved. Rheological studies have been carried out for silica nanoparticles in ionic liquids as the base fluid [101,102]. While rheological measurements allows for testing how the structure of ionic liquids affects the viscoelastic properties of the colloidal dispersions, they do not give access to the dynamics on molecular level. Viscosity of graphene based nanofluids has been to our knowledge for the first time discussed in Ref. [103] for ionic liquid HMIM-BF₄ and in Ref. [100] for ethylene glycol/water mixture. Explanations of the results have been attempted using several hydrodynamical models describing volume concentration dependence of viscosity and failed—viscosity in the nanofluids increases much more significantly with the graphene loading than predicted. On the other hand HMIM-BF₄ containing graphene and carbon nanotubes shows decreasing viscosity compared to the base fluid [103]. Similar effects have been observed for BMIM-C(NC)₂ containing graphene or reduced graphene oxide [104].

Relaxation dispersion profiles for nanofluids are much different than for liquids in bulk and they show numerous peculiar features. Examples of relaxometry data for such systems are shown in Fig. 24.

In the first case (Fig. 24A), for BMIM-BF₄ including carbon nanotubes (of the concentration of about 17% wt) the data differ in the whole frequency range indicating that both the translation and the rotation dynamics of BMIM cations is considerably affected by interactions with the surface of the inserted nanotubes. This is, however, only a first impression and it can be misleading. One could think that the situation is different when carbon nanotubes are replaced by SiO₂ nanoparticles (Fig. 24B). In the high frequency range the ¹H spin-lattice relaxation dispersion profiles overlap. This founding can be treated as an indication that the rotational dynamics (providing the dominating relaxation contribution at high frequencies) remains, in this case, almost unaffected. We wish to stress at this point that as the results are very recent, this is a ‘work in progress’ and therefore we

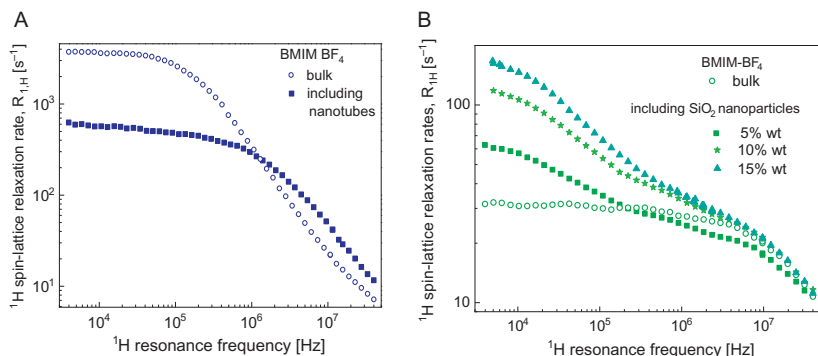


Fig. 24 ^1H spin-lattice relaxation data for (A) a nanofluid composed of BMIM-BF₄ and carbon nanotubes (about 18% wt) at 233 K; corresponding relaxation data for BMIM-BF₄ are shown for comparison and (B) nanofluids composed of BMIM-BF₄ and SiO₂ nanoparticles at 283 K; corresponding relaxation data for BMIM-BF₄ are shown for comparison.

will refrain ourselves from drawing final conclusions. We are, however, of the opinion that independently of the specifics of the system, there are two main contributions to the overall relaxation: from a ‘free’ (bulk) fraction of the liquid and from a fraction being close to the vicinity of surfaces. This statement is far from being obvious—it assumes that dynamics of a part of the solvent remains unaffected (or it is only weakly affected) by the presence of nanoobjects and/or confinement. On the other hand, one could argue that in such systems there is no anymore a fraction of ions (molecules) that behaves like in bulk. This is a fundamental question which should and can be answered by performing detailed, quantitative analysis of the relaxation data. Ideally, the analysis should be supported by NMR diffusometry in order to resolve whether the systems includes pools of ^1H nuclei (or other nuclei relevant for the system) that exhibit translation diffusion occurring on different time scales.

With decreasing temperature the relaxation features of some nanofluids become peculiar. Staying with the example of BMIM-BF₄ containing nanotubes one can make two observations on the basis of Fig. 25. The first one is that in a broad range of frequencies the ^1H and ^{19}F relaxation for the nanoliquid is slower than for the base fluid. This indicates that the translation diffusion of cations and anions in the nanofluid is considerably faster than in the bulk liquid—this result is surprising. The second striking observation is the peculiar shape of the ^1H and ^{19}F relaxation—the relaxation dispersion at very low frequencies becomes very strong. One can consider the following

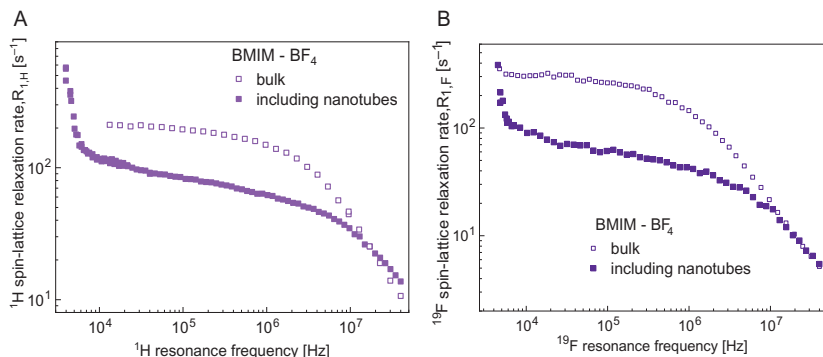


Fig. 25 ^1H (A) and ^{19}F (B) spin-lattice relaxation dispersion profiles for BMIM- BF_4 containing carbon nanotubes compared with BMIM- BF_4 in bulk [46] at 253 K.

explanation. As already said, the dynamics near a surface consists of adsorptions and diffusion steps. In the succession of adsorption steps to the surface of the nanoobject the ion changes its orientation—i.e., it gets adsorbed under a different angle when it again hits the surface; this process is referred to as rotation mediated by translation diffusion [90]. The distribution of times which the molecule spends making the diffusional loops is described by Levy statistic, which means that it is heavy-tailed towards long times. The long time tail of the distribution implies that the molecule eventually escapes to the ‘free’ fraction and one has to wait a long time for it to come back to the vicinity of the surface. In nanofluids the confining walls are finite. A molecule which loses one object (by spending too long time far from it) has to ‘wait’ a long time before it gets the chance to hit a second one. In consequence, the effective reorientational time becomes very long leading to a strong relaxation dispersion at the lowest frequencies.

A much different relaxation (and hence dynamical) scenario is observed for a nanofluid composed of ethylene glycol (a molecular liquid) including graphene sheets (Fig. 26). The relaxation at low frequencies is slower for pure (bulk) ethylene glycol than for the nanofluid. This indicates that the translation diffusion of ethylene glycol molecules becomes faster after adding graphene—this result is rather unexpected.

6.1 Rotation mediated translation diffusion

A mathematical model of motion near surfaces referred to as rotation mediated translation diffusion has been presented in Ref. [94]. When liquids are in contact with porous media there is often a preferential orientation of the

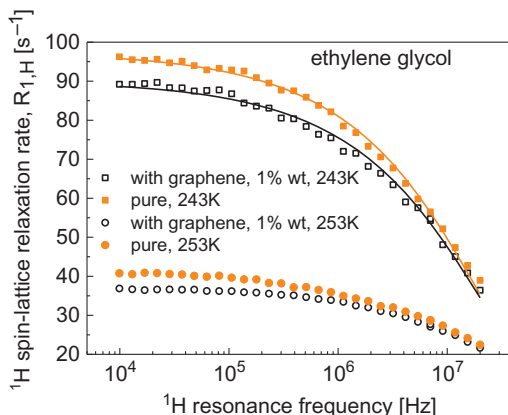


Fig. 26 ^1H spin-lattice relaxation data for a nanofluid formed by ethylene glycol including graphene sheets.

molecules, that might be result of, e.g., creation of a bond between a polar group of a given molecule and a surface site. Thus, translational diffusion along rough surface leads to rotations. This might be the case of hydration shell of biomacromolecules, lipid bilayers, silica nanoparticles in solvent and other types of confinements. Local reorientations of molecules may not lead to decay of the orientation correlation function $G(t)$ down to zero, but there might be a finite residual correlation, $G(\infty) \neq 0$. These non-zero interactions can be further modulated by other dynamic processes of longer timescale. In Ref. [105] a model to describe spin-lattice relaxation due to such 'reorientations mediated by translational diffusion' (RMTD) has been introduced. Characteristic times for the reorientation of a molecule depend on the propagator and on the structure factor $S(q)$ of the confining medium. What is worth to emphasize, RMTD type of relaxation is caused by fluctuations of intra-molecular dipolar (or quadrupolar) interactions of nuclei [105,106]. In the simplest case of homogeneous and isotropic diffusion, the probability density for the displacement $\vec{r}$ in a time t can be treated as a sum of modes characterized by wavenumbers q . The orientational correlation function describing the diffusion associated with a given mode decays with time $G_q(t) \propto \exp\left(-\frac{t}{\tau_q}\right)$ with the decay time defined as $\tau_q = 1/(Dq^2)$. Orientational structure factor and distribution of modes present in the probability function is reflected in the form of the correlation function:

$$G^{RMTD}(t) = \int_0^\infty S(q) \exp(-Dtq^2) dq \quad (34)$$

and corresponding spectral density:

$$J^{RMTD}(\omega) = \int_{q_{min}}^{q_{max}} S(q) \frac{2\tau_q}{1 + \omega^2 \tau_q^2} dq \quad (35)$$

of RMTD process.

One can approximate the structure factor as $S(q) = aq^{-\chi} + b$, with constants a and b ; in low frequency limit the linear term b can be neglected. The exponent χ is connected with the Hurst exponent, α , used to describe the roughness of the surface: $\chi = 1 - \alpha$ [94,107,108]. Then, Eq. (35) can be rewritten as:

$$J^{RMTD}(\omega) \propto \frac{2}{D} \int_{q_{min}}^{q_{max}} \frac{q^{2-\chi}}{q^4 + \left(\frac{\omega}{D}\right)^2} dq = \omega^{-p} \int_{\sqrt{\frac{\omega_{min}}{\omega}}}^{\sqrt{\frac{\omega_{max}}{\omega}}} \frac{z^{3-2p}}{1 + z^4} dz \quad (36)$$

where high and low cut-off frequencies $\omega_{max} = Dq_{min}^2$, $\omega_{min} = Dq_{max}^2$ have been introduced.

Exponent p was defined as $p = \frac{1+\chi}{2}$, thus can be within range (0.5–1). Relaxation rate driven by RMTD process is proportional to ω^{-p} in a range between low and high cut-off frequencies. In case of equipartition of wavenumbers in a given range ($S(q) = \frac{1}{q_{max} - q_{min}} = const.$ for $q_{min} < q < q_{max}$) Eq. (36) simplifies to the form:

$$J^{RMTD}(\omega) = A_{RMTD} \frac{1}{\sqrt{\omega}} \left(f\left(\frac{\omega_{max}}{\omega}\right) - f\left(\frac{\omega_{min}}{\omega}\right) \right) \quad (37)$$

with

$$f(y) = \frac{1}{\pi} \left(\operatorname{atan}\left(\sqrt{2y} + 1\right) + \operatorname{atan}\left(\sqrt{2y} - 1\right) - \operatorname{atanh}\left(\frac{\sqrt{2y}}{y+1}\right) \right) \quad (38)$$

It corresponds to the p exponent value equal to $\frac{1}{2}$ and the relaxation rate being proportional to square root of resonance frequency between ω_{min} and ω_{max} . The amplitude, A_{RMTD} , depends on residual couplings averaged by processes faster than RMTD, structural properties of liquid and on diffusion coefficient and can be approximated as $A_{RMTD} = A_0 D^{-1/2}$, where A_0 is temperature independent constant [109].

Knowing the frequency cut-offs one can estimate the smallest and the longest possible displacements along the surface: $l_{min} = \sqrt{4D/\omega_{max}}$, $l_{max} = \sqrt{4D/\omega_{min}}$ [110].

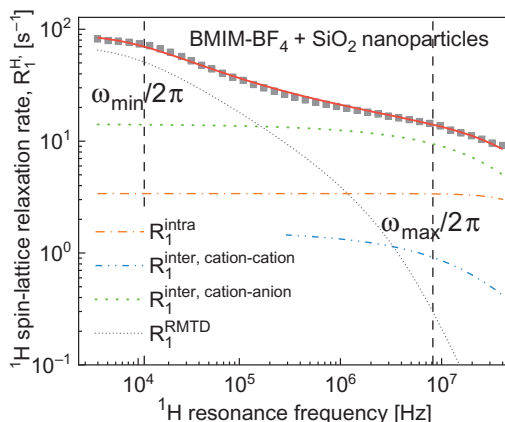


Fig. 27 ^1H spin-lattice relaxation rates, $R_{1H}(\omega_H)$, for ionic liquid BMIM- BF_4 with SiO_2 nanoparticles (10 wt%) recorded at 293 K. Solid line indicates a fit decomposed into contributions related with the translational, rotational and RMTD processes. The fit parameters are: relaxation constant $C = 2.94 \cdot 10^9 \text{ Hz}^2$, $\tau_{\text{rot}} = 0.8 \text{ ns}$, $D_H = 9.2 \cdot 10^{-12} \text{ m}^2/\text{s}$, $D_F = 8.4 \cdot 10^{-12} \text{ m}^2/\text{s}$, $d_{HH} = 2.95 \text{ \AA}$, $d_{HF} = 4.33 \text{ \AA}$, $A_{\text{RMTD}} = 8.75 \cdot 10^3$, $l_{\text{min}} = 9 \text{ \AA}$, $l_{\text{max}} = 253 \text{ \AA}$, $p = 0.5$.

Fig. 27 shows an example of data analysis including a relaxation contribution associated with the rotation mediated translation diffusion process. This example should be treated as an illustration that such an interpretation leads to an agreement with the experimental data, but this is not the only option—different models should be considered as well.



7. Non-exponential relaxation processes

Quite often spin-lattice relaxation processes are expected to be non-exponential. In fact, one can talk about a single-exponential, double-exponential or non-exponential magnetization evolution; attempts to describe the magnetization evolution as, for instance, a three-exponential process are an obvious over-parametrization (unless extremely well justified). The first reason of a non-exponentiality is the composition of the system. An example of two-component systems is a mixture of PS and THF. Fig. 28 shows some ^1H magnetization curves recorded vs time for the mixture at 283 K. The relaxation process is clearly bi-exponential.

Fig. 29 shows the two ^1H spin-lattice relaxation contribution for the whole frequency range up to 40 MHz. In this case one might attempt to straightforwardly explain the bi-exponentiality by saying that the two relaxation components just correspond to the two components of the system.

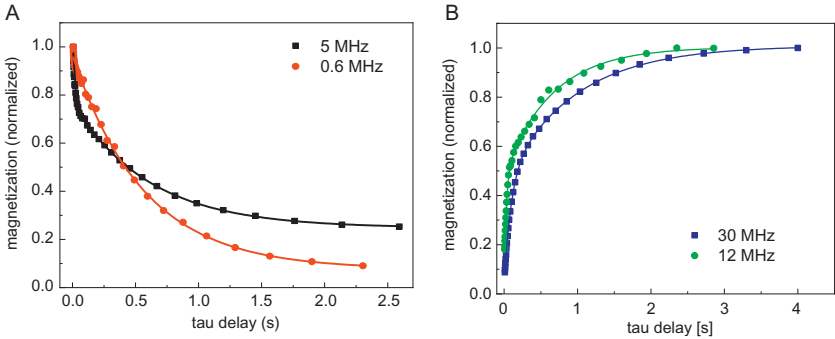


Fig. 28 (A and B). ^1H magnetization curves for a mixture of PS and THF; solid lines—bi-exponential fits.

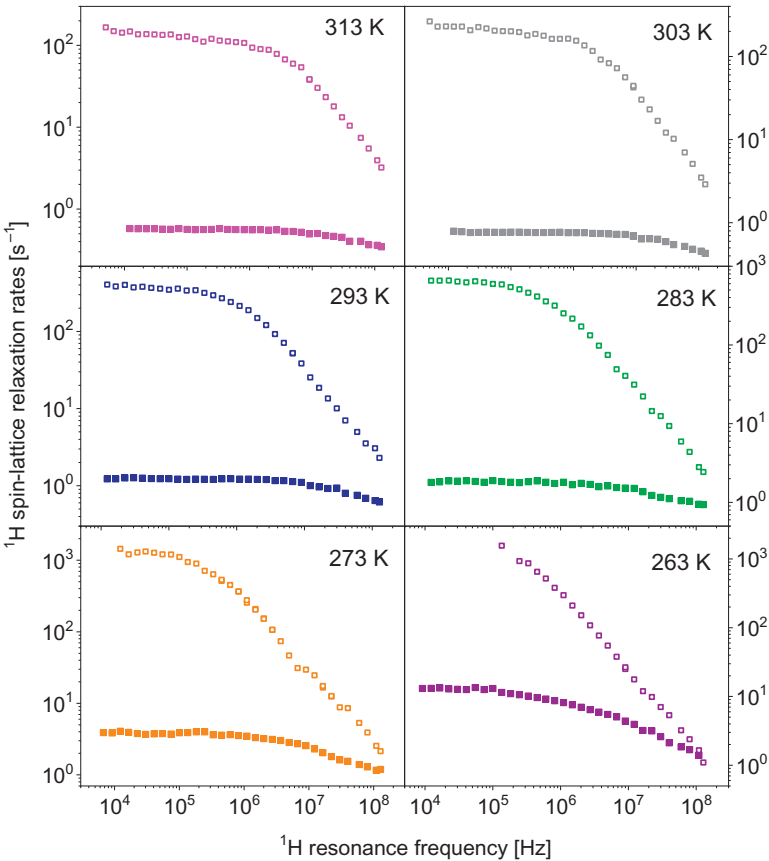


Fig. 29 Bi-exponential ^1H spin-lattice relaxation for THF and PS mixture.

Such an explanation is only partially correct. One may not forget that the PS and THF molecules are not separated—there are mutual ^1H - ^1H dipole-dipole interactions between them that are modulated by the relative translation diffusion of the participating molecules and their relaxation processes are intertwined. This implies that when explaining the origin of the two relaxation components, one should not think only in terms of the molecular subsystems, but rather in terms of different motional processes present in the system. One can expect that THF molecules exhibit faster dynamics than PS molecules. As faster dynamics means (at least at low frequencies and when other parameters are comparable) a slower relaxation, one can attribute the long relaxation component to motional processes involving THF molecules. Thus, the smaller relaxation rate likely stems from three relaxation channels: inter-molecular ^1H - ^1H dipole-dipole interactions between THF molecules modulated by the relative translational diffusion of THF ($2D_{\text{THF}}$), ^1H - ^1H dipole-dipole interactions between THF and PS molecules modulated by the relative translational diffusion of both molecules ($D_{\text{THF}} + D_{\text{PS}}$) and the intra-molecular ^1H - ^1H dipole-dipole interactions modulated by rotational dynamics of THF molecules. Even in the limiting case when the diffusion coefficient of the PS molecules is much slower than of the THF molecules ($D_{\text{PS}} \ll D_{\text{THF}}$), the ratio between the THF-THF and THF-PS relative translation diffusion coefficients, $\frac{2D_{\text{THF}}}{D_{\text{PS}} + D_{\text{THF}}}$, does not exceed factor 2; this is not sufficient to distinguish between the corresponding relaxation contributions.

Nevertheless, one should be aware that the relaxation rate includes a considerable contribution originating from the dipole-dipole interactions between the two kinds of molecules. This cannot be said about the faster relaxation component. The larger relaxation rate is associated with the slow dynamics of the PS molecules including an intra-molecular and an inter-molecular (PS-PS) contributions.

One should, however, be aware that double-exponential relaxation processes can be observed in systems that are homogenous from the structural as well as dynamical viewpoints and the reason of this effect is of quantum-mechanical origin. This happens for systems including two pools of NMR active nuclei that differ in terms of the resonance frequency. One can consider here as examples a system composed of ^1H and ^{19}F nuclei or two pools of ^1H nuclei placed in different chemical surroundings and therefore experiencing different local magnetic fields leading to a shift of their resonance frequencies (referred to as a chemical shift). Let us label the spins

of these two pools of NMR active nuclei as I and S . One should also assume that their energy level structures are entirely determined by their Zeeman interactions; for $I = S = \frac{1}{2}$ this condition is fulfilled for sure. In such a case one can say that their magnetizations, $M_{z,I}(t)$ and $M_{z,S}(t)$, respectively, can be expressed in terms of the $(\langle I_z \rangle(t) - \langle I_z \rangle_{eq})$ and $(\langle S_z \rangle(t) - \langle S_z \rangle_{eq})$ quantities, respectively. The $\langle I_z \rangle(t)$ and $\langle S_z \rangle(t)$ terms denote the averaged values of the I_z and S_z tensor operators, respectively, at a given time-point (they represent the differences in the populations of the available energy levels at this time-point), while the $\langle I_z \rangle_{eq}$ and $\langle S_z \rangle_{eq}$ quantities represent the equilibrium differences in the populations (and hence the equilibrium magnetizations of the I and S spins, respectively). The deviations of the magnetizations from their equilibrium values follow a set of coupled differential equations:

$$\frac{d(\langle I_z \rangle - \langle I_z \rangle_{eq})}{dt} = -R_{II}(\langle I_z \rangle - \langle I_z \rangle_{eq}) - R_{IS}(\langle S_z \rangle - \langle S_z \rangle_{eq}) \quad (39a)$$

$$\frac{d(\langle S_z \rangle - \langle S_z \rangle_{eq})}{dt} = -R_{IS}(\langle I_z \rangle - \langle I_z \rangle_{eq}) - R_{SS}(\langle S_z \rangle - \langle S_z \rangle_{eq}) \quad (39b)$$

where the relaxation rates R_{II} and R_{SS} are referred to as auto-relaxation rates, while R_{IS} and R_{SI} are called cross-relaxation rates. The set of equations (Eqs. (39a) and (39b)) reflects the fact that the transitions of the spins I and S between their energy levels are mutually coupled—they do not occur independently. The solution of this set of equations is bi-exponential and it yields:

$$M_{z,I(S)}(t) = C_{I,(S)}^+ \exp(-R_1^+ t) + C_{I,(S)}^- \exp(-R_1^- t) \quad (40a)$$

where the relaxation coefficients $R_1^\pm$ take the form:

$$R_1^\pm = \frac{1}{2} \left[(R_{II} + R_{SS}) \pm \sqrt{(R_{II} - R_{SS})^2 + 4R_{IS}R_{SI}} \right] \quad (40b)$$

The relaxation rates R_{II} , R_{SS} and $R_{IS} = R_{SI}$ are (for $I = S = \frac{1}{2}$) given as:

$$R_{II} = \frac{1}{2} \left(\frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \right)^2 [J(\omega_I - \omega_S) + 3J(\omega_I) + 6J(\omega_I + \omega_S)] \quad (41a)$$

$$R_{SS} = \frac{1}{2} \left(\frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \right)^2 [J(\omega_I - \omega_S) + 3J(\omega_S) + 6J(\omega_I + \omega_S)] \quad (41b)$$

$$R_{IS} = R_{SI} = \frac{1}{2} \left(\frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \right)^2 [6J(\omega_I + \omega_S) - J(\omega_S - \omega_I)] \quad (41c)$$

Eqs. (41a) and (41b) explain the combinations of the spectral density functions used in Eqs. (31c) and (33c), respectively. The relaxation process of spin I becomes single-exponential when the S spin relaxation is fast ($R_{SS} \gg R_{II}$). Then the S spin sub-system reaches its equilibrium, $\langle S_z \rangle = \langle S_z \rangle_{eq}$, very fast compared to the time evolution of the I spin magnetization and, consequently, Eq. (34) converges to: $\frac{d(\langle I_z \rangle - \langle I_z \rangle_{eq})}{dt} = -R_{II}(\langle I_z \rangle - \langle I_z \rangle_{eq})$, with a single-exponential solution. The fact, that the theory predicts a bi-exponential relaxation process, does not automatically imply that one sees this effect experimentally. First, when the difference between R_{II} and R_{SS} is large the relaxation process becomes single-exponential (as already explained), but when the difference is small one can hardly distinguish between the two relaxation components (the magnetization evolution shows some deviations from a single-exponential function, but they are not sufficient to unambiguously decompose the magnetization into two contributions). Second, even though the relationship between the relaxation rates is appropriate for potentially observing both the relaxation components, one should consider their relative amplitudes. The bi-exponential solution (Eq. 40b) of the set of differential equations Eqs. (39a) and (39b) can be explicitly expressed as:

$$M_{z,I}(t) = C_1 \exp(-R_1^+ t) + C_2 \exp(-R_1^- t) \quad (42a)$$

$$M_{z,S}(t) = C_1 \frac{R_1^+ - R_{II}}{R_{IS}} \exp(-R_1^+ t) + C_2 \frac{R_1^- - R_{SS}}{R_{IS}} \exp(-R_1^- t) \quad (42b)$$

where the constants C_1 and C_2 depend on the initial values of the magnetizations: $M_{z,I(S)}(0)$:

$$C_1 = \frac{R_{IS}M_{z,S}(0) + (R_{SS} - R_1^-)M_{z,I}(0)}{R_1^+ - R_1^- + R_{II} - R_{SS}}, \quad C_2 = M_{z,I}(0) - C_1 \quad (43)$$

One should also note that detecting a two-exponential relaxation process presents a challenge from the experimental viewpoint. One should consider using a logarithmic time scale to fully cover the evolution of the magnetization occurring on a broad time scale. Eventually, it is worth to point out that even though a single-exponential magnetization evolution is observed in the experiment, one should be aware that for complex systems it might just be one of the two components predicted by theory. This means that the resulted relaxation data might be affected by the cross-relaxation terms, R_{IS} , R_{SI} .



8. Nuclear spin relaxation in paramagnetic systems

Let us assume that the S spin represents an electron spin. One can consider here the case of $S = \frac{1}{2}$ (for instance, for nitroxide radicals) or the case of $S \geq 1$ for transition and rare earth metal ions (like Gd or Mn) [111–116]. As electron spin relaxation is very fast compared to nuclear spin relaxation one can safely consider the I spin relaxation as single-exponential. Thus, one can set $R_{1,I}(\omega_I) = R_{II}(\omega_I)$. Then one can generalize Eq. (41a) to the form [62,116–119]:

$$R_{1,I}(\omega_I) = \frac{2}{3} \left(\frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \right)^2 S(S+1) [J(\omega_I - \omega_S) + 3J(\omega_I) + 6J(\omega_I + \omega_S)] \quad (44)$$

As $\omega_S \gg \omega_I$, one can rewrite Eq. (44) in a simpler form:

$$R_{1,I}(\omega_I) = \frac{2}{3} \left(\frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \right)^2 S(S+1) [7J(\omega_S) + 3J(\omega_I)] \quad (45)$$

Eq. (45) can be considered as the beginning of the theory of nuclear spin relaxation induced by dipole-dipole interactions with an electron spin (a paramagnetic centre) that has given rise to the concept of paramagnetic contrast agents used in Magnetic Resonance Imaging (MRI). The increase of the spin-lattice and spin-spin relaxation rates of the surrounding water protons caused by strong $I-S$ dipole-dipole interactions (the ratio between the electron and proton gyromagnetic factors yields 658) is referred to as Paramagnetic Relaxation Enhancement (PRE). Generally, one talks about so called inner-sphere and outer-sphere PRE. The inner-sphere PRE is associated with water molecules forming a coordination sphere around the molecule (particle) carrying the paramagnetic centre (the electron spin). As the water molecules rotate together with the paramagnetic species, one can use the Lorentzian form for the spectral density function; thus:

$$R_{1,I}(\omega_I) = \frac{2}{15} \left(\frac{\mu_0 \gamma_I \gamma_S \hbar}{4\pi r^3} \right)^2 S(S+1) \left[\frac{7\tau_{c,2}}{1 + (\omega_S \tau_{c,2})^2} + \frac{3\tau_{c,1}}{1 + (\omega_I \tau_{c,1})^2} \right] \quad (46)$$

Eq. (46) includes two correlation times, $\tau_{c,1}$ and $\tau_{c,2}$. At this stage one should realize that the paramagnetic centres (electron spins) possess their

own relaxation mechanism, independent of the presence of the neighbouring nuclear spins (^1H nuclei). The mechanism of the electron spin relaxation depends on the system. For transition and rare earth metal ions, the predominant origin of the electron spin relaxation is provided by Zero Field Splitting (ZFS) interactions that can be treated as the electron spin counterpart of the quadrupole coupling for nuclear spins.

The ZFS Hamiltonian takes, in the laboratory (L) frame, the form [120,121]:

$$H_{\text{ZFS}}^{(L)}(S) = \sqrt{\frac{2}{3}} D \sum_{m=-2}^2 (-1)^m U_{-m}^{2(L)} T_m^2(S) \quad (47)$$

where $U_{-n}^{2(L)} = D_{0,-n}^2(\mathbf{\Omega}_{PL}) + \sqrt{\frac{3}{2}} \frac{E}{D} [D_{-2,2}^2(\mathbf{\Omega}_{PL}) + D_{2,2}^2(\mathbf{\Omega}_{PL})]$. The parameters D and E describe the axial and rhombic components of the ZFS tensor; (P) denotes the principal axis system of the ZFS tensor, $\mathbf{\Omega}_{PL}$ describes the orientation of this frame with respect to the laboratory frame. Comparing the expressions for $U_m^{2(P)}$ for ZFS interactions and $V_m^{2(P)}$ for quadrupole interactions ($V_0^{2(P)} = 1$, $V_{\pm 1}^{2(P)} = 0$, $V_{\pm 2}^{2(P)} = \eta/\sqrt{6}$) one can see the analogy: $D \rightarrow \frac{3}{4} \frac{a_Q}{S(2S-1)}$ and $E \rightarrow \frac{1}{4} \eta \frac{a_Q}{S(2S-1)}$.

The correlation times, $\tau_{c,1}$ and $\tau_{c,2}$, describe the effective fluctuations of the $I-S$ dipole-dipole coupling. The fluctuations originate from the rotational dynamics of the paramagnetic species including the water coordination sphere described by the rotational correlation time τ_{rot} , the exchange dynamics between the coordination sphere and bulk water described by the exchange life time, τ_{ex} , and the electron spin relaxation. This implies that the correlation times can be expressed as: $\frac{1}{\tau_{c,1}} = \frac{1}{\tau_{rot}} + \frac{1}{\tau_{ex}} + R_{1,S}$ and $\frac{1}{\tau_{c,2}} = \frac{1}{\tau_{rot}} + \frac{1}{\tau_{ex}} + R_{2,S}$, where $R_{1,S}$ and $R_{2,S}$ denote the electron spin spin-lattice and spin-spin relaxation rates, respectively. The relaxation rates are also dependent of the magnetic field; they can be expressed as:

$$R_{1,S} = \left(\frac{1}{3} D^2 + E^2 \right) [J(\omega_S) + 4J(2\omega_S)] \quad (48a)$$

$$R_{2,S} = \left(\frac{1}{3} D^2 + E^2 \right) [3J(0) + 5J(\omega_S) + 2J(2\omega_S)] \quad (48b)$$

Assuming that the stochastic fluctuations of the ZFS tensor can be described by an exponential correlation function (this is far from being obvious) with a correlation time, τ_{ZFS} , one can insert to Eqs. 48a and 48b the Lorentzian form of the spectral density: $J(\omega) = \frac{1}{5} \frac{\tau_{\text{ZFS}}}{1 + (\omega\tau_{\text{ZFS}})^2}$.

Using this model one can capture the essential features of PRE provided several conditions are fulfilled. First of all, one can clearly deduce from the form of Eqs. (46), (48a) and (48b) that the description is based on the assumption that the energy level structure of the electron spin is solely determined by its Zeeman coupling. This can be true in two situations: the Zeeman interaction is much larger than the ZFS interaction (this is possible at high magnetic fields) or the rotational motion of the paramagnetic species is fast enough to fulfil the relationship: $\omega_{ZFS}\tau_{rot} \ll 1$, where ω_{ZFS} is the amplitude of the ZFS interaction in angular frequency units. In the case of fast rotation the ZFS interaction acts as a source of the electron spin relaxation; it does not contribute to the energy level structure of the electron spin. The scenario becomes much more complex for slow rotation. Actually, from the viewpoint of the efficiency of paramagnetic contrast agents slowly rotating systems are preferred as then the PRE effect becomes larger. In such a case the electron spin energy level structure is determined by a superposition of its Zeeman and ZFS interactions. We have already considered an analogous situation for quadrupole nuclei. Then, however, one should understand that the ZFS tensor can be split into two parts: a part fixed in the molecule and hence rotating together with the paramagnetic species and a part already fluctuating in the molecule due to, for instance, its internal dynamics. The second part always acts as a mechanism of the electron spin relaxation, independently of the time scale of the molecular tumbling. However, when the energy level structure of the electron spin is non-Zeeman (and this takes place for slow rotational dynamics of the paramagnetic species) one has to consider several electron spin relaxation rates and their frequency dependences are much more complex than those given by Eqs. (48a) and (48b).

The spin-lattice relaxation rate, $\widetilde{R}_{1,I}(\omega_I)$, measured for solutions of paramagnetic molecules is given as:

$$\widetilde{R}_{1,I}(\omega_I) = \frac{Pq}{(R_{1,I}(\omega_I))^{-1} + \tau_{ex}} + R_{1,I}^{outer}(\omega_I) \quad 49$$

where P denotes the mole fraction of water in the bound position, while q is the coordination number; $R_{1,I}^{outer}(\omega_I)$ represents the outer sphere PRE associated with translation diffusion of the solvent (water) molecules.

Although one might see some analogies between the PRE and QRE effects, this impression is highly misleading. In fact, the only analogy is the mathematical formalisms of the ZFS and quadrupole interactions in terms of their Hamiltonians. One should realize that the physical origins

of PRE and QRE is different. The PRE effect stems from strong nuclear spin–electron spin dipole–dipole coupling. Consequently, an enhancement of the nuclear (let us focus on ^1H for simplicity) is present at any magnetic field (resonance frequency). Due to the complex electron spin dynamics (relaxation) the level of the enhancement depends on the resonance frequency, but the effect is always present. This does not apply to QRE. ^1H nuclei possess the largest gyromagnetic factor among all nuclei. This implies that I – S dipole–dipole interactions between ^1H and quadrupole nuclei will rather not exceed ^1H – ^1H dipole–dipole interactions (unless the inter-spin distance for the I – S pair is considerably shorter). The essence of QRE effects lies in the matching of the transition frequencies between the energy levels of the quadrupole nucleus and the ^1H resonance frequency. Such a matching is impossible for electron spins due to their strong Zeeman coupling—the electron spin transition frequencies are always much larger than the ^1H resonance frequency. Nevertheless, it is well-known that very often one observes a very significant maximum of ^1H relaxation rates for the case of PRE; an example is given in Fig. 30.

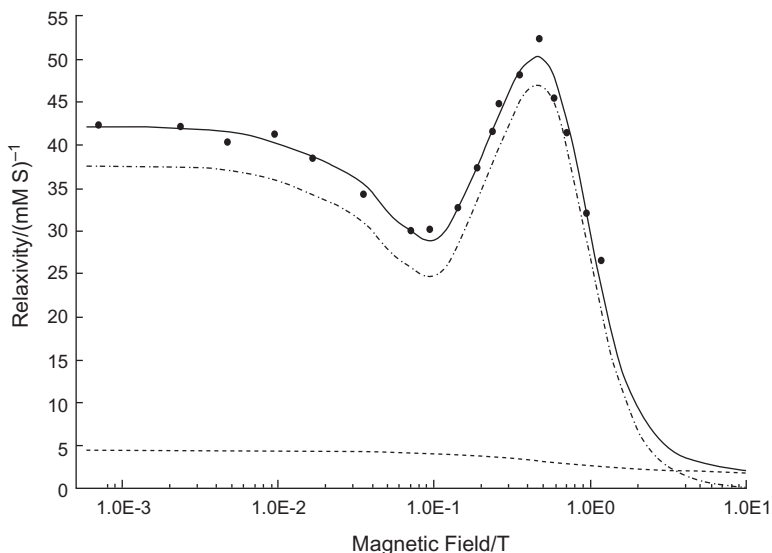


Fig. 30 Experimental and calculated NMRD profiles for GdDOTA(BOM) in aqueous solution at 298 K. The dashed-dotted line corresponds to the inner-sphere contribution, while the outer sphere contribution is presented as the dashed line. Reprinted from Ref. D. Kruk, T. Nilsson, J. Kowalewski, *Nuclear spin relaxation in paramagnetic systems with zero-field splitting and arbitrary electron spin*, *Phys. Chem. Chem. Phys.* 3 (2001) 4907–4917, with permission.

The maximum results from the frequency dependencies of the electron spin relaxation rates. One can say that the fluctuations of the $I-S$ dipole-dipole coupling are described by frequency dependent correlation times. When the electron spin relaxation is slower than the rotational motion or the exchange dynamics, its contribution to the modulations of the $I-S$ dipole-dipole interaction becomes negligible and, consequently, the PRE does not exhibit any maximum. For the case of QRE, the quadrupole spin relaxation is rather slow and it can hardly compete with the other sources of the modulations of the dipole-dipole interaction. Eventually, it is worth to point out that when the condition $(R_{1,I}(\omega_I))^{-1} \gg \tau_{ex}$ is fulfilled, τ_{ex} can be neglected in the denominator of Eq. (49). This is not always the case of PRE, but it applies to QRE effects.



9. Superparamagnetic systems

As far as nanoparticles are concerned their large magnetic moments provide a great advantage leading to very strong dipole-dipole interactions with protons. Superparamagnetic nanoparticles with appropriate surface chemistry can be used for numerous in vivo applications, such as MRI contrast enhancement, tissue repair, immunoassay, detoxification of biological fluids, hyperthermia, drug delivery, and cell separation. All of these biomedical applications require that the nanoparticles have high magnetization values, a size smaller than 100 nm (the sizes should be comparable to the largest biological molecules, such as enzymes, receptors or antibodies), and a narrow particle size distribution. These applications also need specific surface coating of the magnetic particles, to ensure biocompatibility and surface functionalization with biological objects such as antibodies to deliver nanoparticle to the target. Superparamagnetic nanoparticles often increase not only the ^1H spin-lattice, but also the spin-spin relaxation rate. This is caused by the fact that according to spin relaxation theories, the spin-spin relaxation rate depends on a spectral density function taken at zero magnetic field, and this contribution ‘survives’ when the other disappear with increasing magnetic field due to slow rotational dynamics of the nanoparticles (they are large compared to paramagnetic complexes) and relatively slow fluctuations of the magnetic moment (this effect corresponds to electronic relaxation for paramagnetic contrast agents). The most common nanoparticles are composed of magnetite (Fe_3O_4) or maghemite ($\gamma\text{-Fe}_3\text{O}_4$).

^1H relaxation in water solutions of magnetic nanoparticles is caused by magnetic dipole-dipole interactions between the magnetic moment of the nanoparticle and the proton magnetic moment. The magnetic moment of the nanoparticle has two components: a mean (average) part which is parallel (longitudinal) to the external magnetic field and does not fluctuate in time, and a component that fluctuates around this average value. Both components depend on the applied magnetic field. The mean component increases, according to the Langevin function, from zero, at zero magnetic field, towards a saturation value at high field [122–129]. As the mean component grows the fluctuating part of the longitudinal magnetic component gets lower, i.e., it reaches maximum at zero field and decreases towards zero with increasing field. In consequence, the measured ^1H relaxation rate is a sum of two relaxation contributions. The first contribution is associated with the mean component of the magnetic moment. It is referred to as Curie relaxation and dominates at higher magnetic fields. The second one, associated with the fluctuating component of the nanoparticle magnetization is responsible for the low field relaxation [122–129]. To quantitatively describe the relaxation contributions one has to identify the sources of time fluctuations of the dipole-dipole interactions between the magnetic moments of the nanoparticle and of the solvent. In the theoretical descriptions presented so far in the literature [122–129] it has been assumed that the dipolar interactions associated with the mean magnetization component (Curie relaxation) fluctuate in time due to translational diffusion of protons (solvent molecules), as schematically shown in Fig. 31.

The theory of ^1H spin-lattice relaxation induced by superparamagnetic nanoparticles in solution has been recapitulated in Ref. [130]. The overall spin-lattice relaxation rate includes two contributions: one of them is

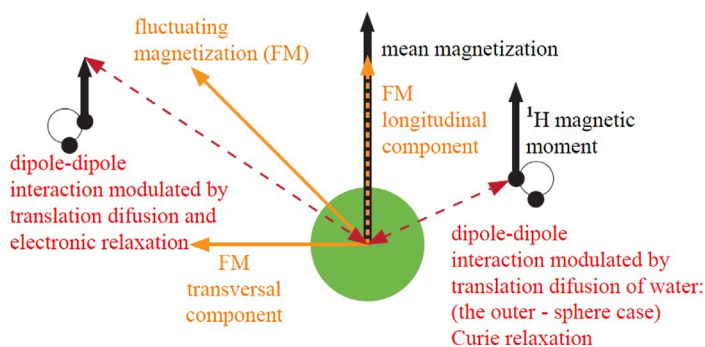


Fig. 31 Schematic view of the mechanism of ^1H relaxation caused by magnetic nanoparticles.

associated with the permanent (fixed in the nanoparticle) part of the magnetic moment, $R_{1,H}^C(\omega_H)$, while the second one, $R_{1,H}^F(\omega_H)$, is associated with the fluctuating part of the magnetic moment:

$$R_{1,H}(\omega_H) = R_{1,H}^C(\omega_H) + R_{1,H}^F(\omega_H) \quad (50)$$

where:

$$R_1^C(\omega_H) = \frac{432}{5} \left(\frac{\mu_0}{4\pi} \gamma_H \gamma_S \hbar \right)^2 \frac{N}{d^3} S^2 \left(\coth \frac{S\hbar\omega_S}{kT} - \frac{1}{2S} \coth \frac{\hbar\omega_S}{2kT} \right)^2 \times \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \frac{u^2 \tau_{trans}}{u^4 + (\omega \tau_{trans})^2} du \quad (51)$$

and

$$R_{1,H}^F(\omega_H) = R_{1,H}^{F(n)}(\omega_H) + R_{1,H}^{F(e)}(\omega_H) \quad (52)$$

where:

$$R_1^{F(n)}(\omega_H) = \frac{432}{5} \left(\frac{\mu_0}{4\pi} \gamma_H \gamma_S \hbar \right)^2 \frac{N}{d^3} \times \left[S(S+1) - S \coth \frac{\hbar\omega_S}{2kT} \left(\coth \frac{S\hbar\omega_S}{kT} - \frac{1}{2S} \coth \frac{\hbar\omega_S}{2kT} \right) - S^2 \left(\coth \frac{S\hbar\omega_S}{kT} - \frac{1}{2S} \coth \frac{\hbar\omega_S}{2kT} \right)^2 \right] \times \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \frac{(u^2 + \tau_{trans} R_{1e}) \tau_{trans}}{(u^2 + \tau_{trans} R_{1e})^2 + (\omega_H \tau_{trans})^2} du \quad (53)$$

while:

$$R_1^{F(e)}(\omega_H) = \frac{1008}{5} \left(\frac{\mu_0}{4\pi} \gamma_H \gamma_S \hbar \right)^2 \frac{N}{d^3} S \coth \frac{\hbar\omega_S}{2kT} \left(\coth \frac{S\hbar\omega_S}{kT} - \frac{1}{2S} \coth \frac{\hbar\omega_S}{2kT} \right) \times \int_0^\infty \frac{u^2}{81 + 9u^2 - 2u^4 + u^6} \frac{(u^2 + \tau_{trans} R_{2e}) \tau_{trans}}{(u^2 + \tau_{trans} R_{1e})^2 + (\omega_S \tau_{trans})^2} du \quad (54)$$

Fig. 32A and B shows ^1H spin-lattice relaxation data for water solution of 5 nm (diameter) and hexane solution of 10 nm spherical iron oxide nanoparticles.

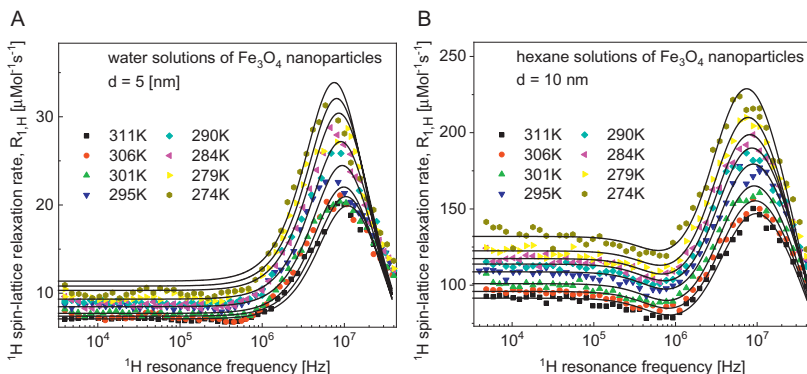


Fig. 32 (A and B) ^1H spin-lattice relaxation dispersion for (A) water solutions of 5 nm iron oxide nanoparticles and (B) hexane solution of 10 nm iron oxide nanoparticles.

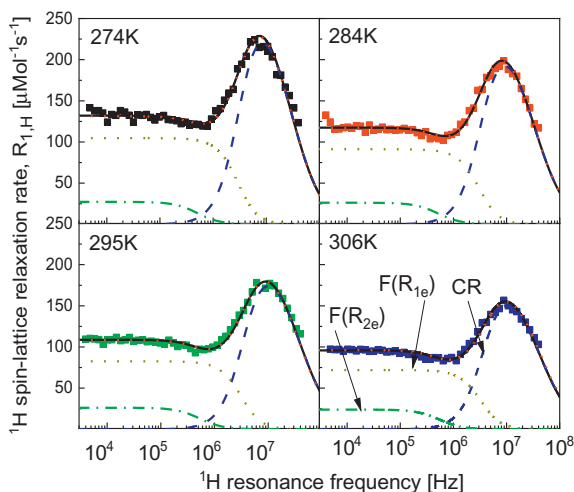


Fig. 33 ^1H relaxation dispersion for hexane solution of 10 nm iron oxide nanoparticles. The overall relaxation process has been decomposed into the Curie spin relaxation (CR) and the relaxation contributions associated with the fluctuating part of the magnetic moment including the electron spin-lattice and spin-spin relaxation rates, $F(R_{1e})$ and $F(R_{2e})$, respectively.

Fig. 33 shows a decomposition of some of the relaxation data for the 10 nm nanoparticles into the Curie relaxation and the component attributed to the fluctuating part of the magnetic moment which has been, in turn, decomposed into the terms associated with electron spin-lattice and spin-spin relaxation.



10. Different representations of relaxation data

Eqs. (2) and (41a) describing I spin-lattice relaxation rates (let's denote them as $R_{1,I}^H(\omega_I)$ and $R_{1,I}^H(\omega_I)$) caused by dipole-dipole interactions between equivalent and non-equivalent nuclei, respectively, can be rewritten as (assuming Lorentzian spectral density functions):

$$R_{1,I}^H(\omega_I) = C_{DD}^H \left[\frac{\tau_c}{1 + (\omega_I \tau_c)^2} + \frac{4\tau_c}{1 + (2\omega_I \tau_c)^2} \right] \quad (55a)$$

$$R_{1,I}^{IS}(\omega_I) = C_{DD}^{IS} \left[\frac{\tau_c}{1 + ((\omega_I - \omega_S) \tau_c)^2} + \frac{3\tau_c}{1 + (\omega_I \tau_c)^2} + \frac{6\tau_c}{1 + ((\omega_I + \omega_S) \tau_c)^2} \right] \quad (55b)$$

where the quantities C_{DD}^H and C_{DD}^{IS} are referred as dipolar relaxation constants. Then one can multiply both sides of Eqs. (55a) and (55b) by the resonance frequency ω_I , yielding:

$$\omega_I R_{1,I}^H(\omega_I) = \chi^H = C_{DD}^H \left[\frac{\omega_I \tau_c}{1 + (\omega_I \tau_c)^2} + \frac{2(2\omega_I \tau_c)}{1 + (2\omega_I \tau_c)^2} \right] \quad (56a)$$

$$\begin{aligned} \omega_I R_{1,I}^{IS}(\omega_I) &= \chi^{IS} \\ &= C_{DD}^{IS} \left[\frac{\omega_I \tau_c}{1 + ((\omega_I - \omega_S) \tau_c)^2} + \frac{3\omega_I \tau_c}{1 + (\omega_I \tau_c)^2} + \frac{6\omega_I \tau_c}{1 + ((\omega_I + \omega_S) \tau_c)^2} \right] \end{aligned} \quad (56b)$$

Substituting $\omega_I \tau_c = x$ one can clearly see that χ^H in Eq. (56a) becomes just a function of the x variable: $\chi^H(x)$. At this stage one can introduce a susceptibility function: $\epsilon(x) = \frac{x}{1+x^2}$; the function has Debye form. Analogously one can construct susceptibility functions of different forms, for instance, Cole–Davidson form by multiplying the spectral density of Eq. (11) by ω . Fig. 34A shows the result of multiplying the simulated relaxation data (Fig. 1) by ω . The peak represent a sum of two Debye functions: $\chi^H(x) = C_{DD}^H[\epsilon(x) + 2\epsilon(2x)]$. As only the product $\omega_I \tau_c$ is of relevance (not the values of ω_I and τ_c separately) one can shift the $\chi^H(x)$ curves unless an overlapping is reached (Fig. 34B). This effect is referred to as frequency-temperature superposition.

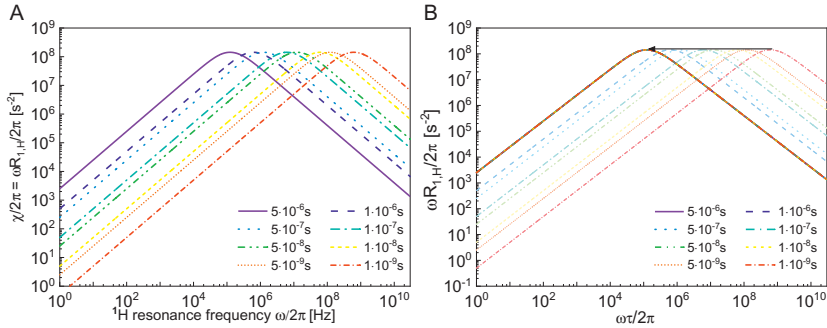


Fig. 34 Susceptibility representation of relaxation data: (A) $\chi''(\omega/2\pi)$ dependence obtained by multiplication of relaxation rates from Fig. 1 by resonance frequency, (B) effect of horizontal shift of susceptibility curves (equivalent to multiplication of frequency by a constant factor).

For real experimental data such an overlapping can be reached (after multiplying the experimental relaxation data collected at different temperatures by the resonance frequency and appropriately shifting the resulted curves) if the only change in the dynamics of the system with temperature is limited to a change of the correlation time. In the case when the mechanism of the motion or other parameters, like for instance the amplitude of vibrational dynamics, change as well, the concept breaks down. An example of the susceptibility representation of ^1H spin-lattice relaxation dispersion profiles and the frequency-temperature superposition is shown in Fig. 35 (unfortunately, for this data set only the left flank of the peak is visible). For comparison, corresponding dielectric spectra for the water solutions of carboxymethylcellulose are shown in Fig. 36.

One should note that the dielectric peak observed in the frequency range of about 10–100 KHz is not present in the relaxation dispersion profiles. This observation is a part of a general question about the complementarity of NMR relaxometry and Dielectric Spectroscopy and the types and mechanisms of dynamical processes probed by both methods.

When the relaxation process is caused by dipole-dipole interactions between non-equivalent nuclei, one can modify Eq. (56b) to the form:

$$\begin{aligned} \omega_I R_{1,I}^{IS}(\omega_I) = \chi^{IS} = C_{DD}^{IS} & \left[\left(\frac{\omega_I}{\omega_I - \omega_s} \right) \frac{(\omega_I - \omega_s)\tau_c}{1 + ((\omega_I - \omega_s)\tau_c)^2} + \frac{3\omega_I\tau_c}{1 + (\omega_I\tau_c)^2} \right. \\ & \left. + \left(\frac{\omega_I}{\omega_I + \omega_s} \right) \frac{6(\omega_I + \omega_s)\tau_c}{1 + ((\omega_I + \omega_s)\tau_c)^2} \right] \end{aligned} \quad (57)$$

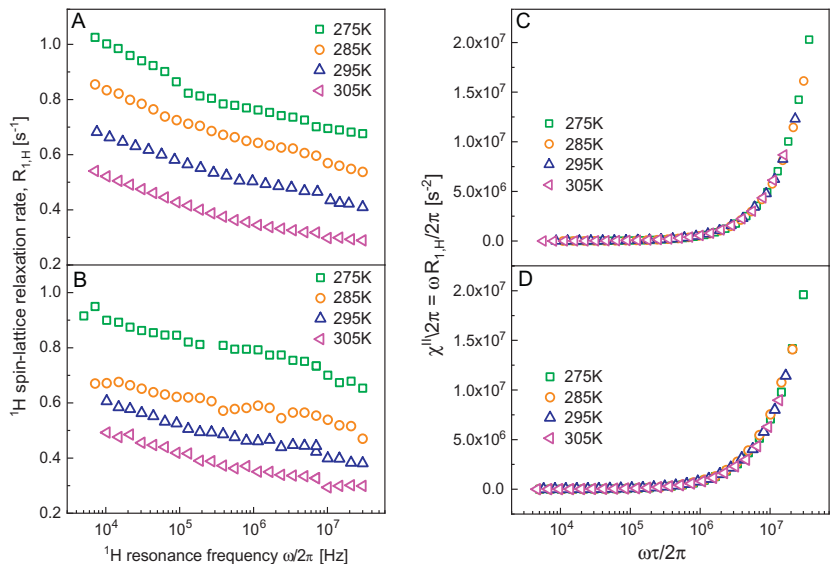


Fig. 35 ^1H spin-lattice relaxation dispersion profiles for water solutions of carboxymethylcellulose (A) 90 kD, (B) 700 kD after subtracting the relaxation contribution from bulk water. The same data in the susceptibility representation: (C) 90 kD, (D) 700 kD.

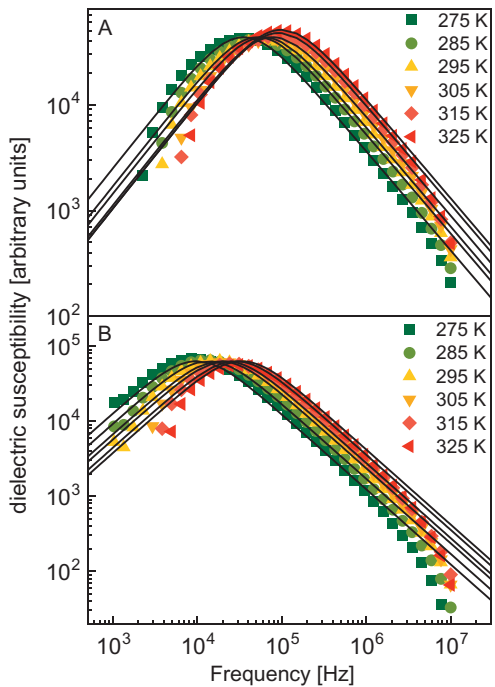


Fig. 36 Dielectric spectra for water solutions of carboxymethylcellulose (A) 90 kD, (B) 700 kD.

Then, using the ratio between the gyromagnetic factors, $\varsigma = \frac{\gamma_s}{\gamma_I}$, one can further re-formulate Eq. (57):

$$\chi^{IS}(x) = C_{DD}^{IS} \left[\left(\frac{1}{1-\varsigma} \right) \epsilon((1-\varsigma)x) + 3\epsilon(x) + 6 \left(\frac{1}{1+\varsigma} \right) \epsilon((1+\varsigma)x) \right] \quad (58)$$

This expression shows that before drawing any conclusions regarding the time scales of dynamical processes on the basis of spin-lattice relaxation dispersion data shown in the susceptibility representation by just looking at the positions of the peaks, one has to carefully consider the quantum-mechanical scenario of the relaxation process as it determines the combinations of the spectral density (susceptibility) functions.



11. Conclusions

In this chapter several examples of applications of NMR relaxometry to complex, liquid and solid systems have been presented. The common denominator of all the studies is the fact that spin relaxation is a quantum-mechanical phenomenon. This should not be treated as an obstacle, but as a challenge. The information about molecular and ionic dynamics can be extracted from NMR relaxometry data only when appropriate theoretical models are available. This fact may not be ignored because applying very simple relaxation formulae and neglecting their validity conditions can lead to a total misinterpretation of the experimental data. The misinterpretation might be of very serious nature; for instance, one might conclude that a system undergoes a heterogeneous motion while the reality is different (for instance, the motion turns out to be homogenous when all relevant relaxation pathways are included) or one might attribute purely quantum-mechanical effects to dynamical processes that, in reality, do not exist. The theoretical effort that is needed to fully profit from the unique potential of NMR relaxometry is, however, rewarding—one can hardly obtain detailed information about molecular motion (independently of its mechanism) on the time scale between ms and ns by other methods. Moreover, thanks to the quantum-mechanical scenario of spin relaxation, the relaxation processes show ‘peculiarities’ like PRE or QRE that can be exploited not only in medical diagnosis, but also in material science, chemistry, biology and food science.

Acknowledgement

The work has partially been supported by National Science Centre, Poland, under the grant number: UMO-2017/25/B/ST5/02348. COST Action CA15209 (European Network on NMR Relaxometry) is greatly acknowledged.

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